An agent based model of Brenner´s theory of transition

Abstract

The results of the Brenner Debate produced an enhanced theory of transition from feudalism to capitalism based in rural 17^{th} Century England. A key tenant of this school of thought is the role of "improvement" as both a motor in the divorce between peasants and their means of subsistence, and a consequence of newly created market imperatives both landlords and tenants were now subject too. To test the role of improvement ideology in the transition an agent based model is constructed. This incorporates tenants, landlords, ecological deterioration and rental markets. The critical role of rental market pressure is modelled through a transition from customary rent to leasehold rents.

1 Introduction

According to [Comninel, 2000, p.10] feudalism may be defined as a mode of production in which the ruling class extracts surplus value from direct producers via non-economic means, often political and under the threat of force. In contrast, capitalism extracts surplus value from labourers, who are not direct producers and do not posses the means of production, via economic means. Capitalists must re-invest the extracted surplus value, in order to reproduce their class position, to keep up with increasing productivity. Through an analysis of the transition from one class structure to another, one might hope to gain insights into common identifiers which are associated with the collapse of a mode of production. Improvement ideology emerged in the mid 16^{th} Century composed of both practical methods to increase the output of a given piece of land, and as a method through which to better the condition of the English peoples, as argued by Captain Walter Blith.

1.1 Marx on the transition and primitive accumulation

Marx argued against the inevitability of capitalism, comparing it to the "original sin" in theology [Marx, 1894, p.507]. Where capitalism's "origin is supposed to be explained when it is told as an anecdote of the past" [Marx, 1894, p.507]. Marx instead produced a theory based off the end result and its defining characteristics of wage co-modification and the separation of the means of production from wage-employees. For Marx the transformation of goods into capital requires a social transformation in the relations between the producer and the ruling class. This process Marx labels Primitive Accumulation. He identified the "expropriation of the self-supporting peasants" as a factor but also "the destruction of rural domestic industry" [Marx, 1894, p.531]. The former of which is the first step in the formation of capital, drawing a distinction between feudal wealth and capital. Marx states "what enables money-wealth to become capital is the encounter, on one side, with free workers; and on the other side, with the necessaries and materials etc, which previously were in one way or another the property of the masses who have now become object-less, and are also free and purchasable" [Marx, 2005]. Marx identifies this process of dispossession as having two components, firstly the "forcible driving of the peasantry from [their] land" and secondly a "usurpation of the common lands" [Marx, 1894, p.514]. With both factors Marx uses Francis Bacon's "History of the Reign of King Henry VII" to pinpoint historical legislation off the back of which much of this dispossession was carried out. For Marx these tendencies contributed to a final state of a "degraded and almost servile condition of the mass of the people, the transformation of them into mercenaries, and of their means of labour into capital" [Marx, 1894, p.511]. This degradation of the peasantry can be seen in the collapse of the independent yeoman who was replaced by tenants who depended on yearly leases. Producing "a servile rabble dependent on the pleasure of the landlords" [Marx, 1894, p.511]. This available labour was now "transformed into material elements of variable capital" [Marx, 1894, p.530] thereby introducing the final components for the transition, " the peasant, expropriated and cast adrift, must buy their value in the form of wages, from his new master, the industrial capitalist" [Marx, 1894, p.530]. For Marx it is a combination of social relations and material conditions that produces the banality of capitalist economic logic resulting in "the subjugation of the labourer to the capitalist" [Marx, 1894, p.523]. In his analysis Marx had a very limited mention of improvement language. Suggesting the "revolution in the conditions of landed property was accompanied by improved methods of culture, greater co-operation, concentration of the means of production" [Marx, 1894, p.530]. This lack of involvement of improvement within Marx's theory of transition could in part be explained by a lack of access to many of the primary sources used by the likes of Brenner and Woods in constructing their own theories. But also speaks to the fact that Marx was first and foremost a philosopher not a historian.

1.2 Brenner's theory of transition

With the publication of "Agrarian Class Structure and Economic Development in Pre-Industrial Europe" in 1976 Robert Brenner produced a crucial advance in the causes of the very first case study of the transition from feudalism to capitalism. It is in this very word "transition" that Brenner argued even Marxist historians such as Paul Sweezy or Maurice Dobb had been led astray [Brenner, 1977, p.41-53], emphasising that they were assuming an a priori existence of an embryonic capitalism within feudal society.Several theories of transition may be grouped as commercial models. The keystone of these theories is that capitalism arose as the barriers for its development were removed [Wood, 2002, p.12]. Crucially, the driving factor behind this development was man's "propensity to truck, barter, and exchange" [Smith and Stewart, 1963]. This force led to the division of labour and ultimately industrial capitalism. Brenner and Wood's fundamental critique of this approach is that it assumes a nascent agricultural or industrial capitalism within other modes of production, such that there is no real transition and capitalism was inevitable [Wood, 2002, p.51]. Brenner's theory sought internal conflicts within feudalism, as opposed to an external impetus, which might produce its collapse. Wood offers an analysis that " it [was] a matter of lords and peasants, in certain specific conditions peculiar to England, involuntarily setting in train a capitalist dynamic while acting in class conflict with each other to reproduce themselves as they were" [Wood, 2002, p.52]. This focus on the class conflict nature of the transition is what distinguished Brenner and Wood's theory from other contemporary Political Marxists. Brenner identifies the internal logic of feudalism that inhibits surplus reinvestment. For the feudal lord it was often easier to find new avenues to "squeeze" [Brenner, 1976, p.74] out more absolute surplus value from their subjects instead of increasing their labour efficiency. Due to the fact that a lord's power was directly correlated to size of the population under his control this lead to a focus on labour immobility, and maximising the ease with which he could extract surplus value from his peasants by non-economic means. Resulting in restrictions on land mobility and "unfree peasants [unable] to convey their land to other peasants without the lord's permission" [Brenner, 1976, p.50]. The centuries previous to the 17^{th} Century were characterised by the end of serfdom in England, resulting in peasants whose lords extracted surplus value partly through the rent of their land. These were customary rents that decreased in value for the lord over time due to inflation. This loss in income revenue was counteracted through the imposition of arbitrary fines on peasants wherever land was sold or inherited. However over the 15^{th} and 16^{th} centuries landlords successfully crushed peasant land rights in order to combat this decreasing power. This decay in freehold rights meant that lord could slowly accumulate large demesne holdings, which could be leased out to larger tenants [Hoyle, 1990, p.7]. Who in turn sub-let to smaller farmers or employed wage labourers. This new wave of larger tenancies brought two new sources of market imperatives, firstly the lord had to keep his rents competitive as he was now trying to undercut the rest of the domestic tenancy market, given that the larger tenant had the ability to move away.[1] Secondly the tenant had to compete on a goods market, requiring them to sell the products of their husbandry in order to make enough profit to fulfil market dependant rent. This required a shift from production for subsistence to production for sale. These two forces resulted in a new weakly symbiotic relation between lord and capitalist tenant in which both were driven to re-invest surplus value back into their land, thereby improving agricultural yields and transforming feudal wealth into capital. This established a new set of laws of motion in which the lord and tenant were compelled to improve their agricultural yields in order to self-replicate

[1] **Flagged for revision.** This sentence has the competitive pressure running from lords bidding against one another to retain mobile tenants. That is Postan's supply-and-demand account, which Brenner cites only in order to reject it: "the lords and the employers found that the most effective way of retaining labour was to pay higher wages, just as the most effective way of retaining tenants was to lower rents and release servile obligations" [Brenner, 1976, p.39, n.17, quoting Postan]. For the sixteenth century Brenner has the pressure running the other way, since population was rising: "competition for land induces the peasantry to accept a serious degradation of their personal/tenurial status in order to hold on to their land" [Brenner, 1976, p.38], and lords "could always get replacements, quite often indeed on better terms" [Brenner, 1976, p.44]. The market imperative on the lord's side is better stated as the opportunity to let to the highest bidder rather than the necessity of undercutting rivals – see §4.7.

their class condition. Brenner claims that "agricultural improvement not only made it possible for an ever greater proportion of the population to leave the land to enter industry" [Brenner, 1976, p.67] but also mean that a larger proportion of the domestic population was no longer involved in agriculture, freeing peasants to become the wage labour required for "England's continued industrial growth" [Brenner, 1976, p.67] through the 17^{th} Century.

2 Where the debate now stands

Before the questions, a word on the terrain, because it has moved. The exchange that followed Brenner's 1976 intervention [Aston and Philpin, 1985] was fought against demographic and commercial accounts, and §1.1–§1.2 above reconstruct it on those terms. Neither is where the argument now is. Two later fronts have opened, and it is against these, not against Postan [Postan and Hatcher, 1978], that a contribution has to be positioned.

The first is *empirical*, and concerns whether English custom in fact decayed the way Brenner says. Whittle's Norfolk evidence finds an active peasant land market and customary tenure considerably more secure than the account requires, with capitalist relations emerging without wholesale dispossession [Whittle, 2000]; Bailey relocates the decline of serfdom in law and institutional change rather than in class struggle [Bailey, 2014]; Dyer makes the transition longer and earlier than the sixteenth–seventeenth-century window [Dyer, 2005]; Shaw-Taylor's quantitative work pushes proletarianisation later and reduces enclosure's share of it [Shaw-Taylor, 2012]; and Allen locates the productivity gains in yeomen and owner-occupiers rather than in large capitalist tenants on consolidated farms [Allen, 1992] – which, if right, misplaces the causal core of the improvement story. Dimmock's defence engages this evidence directly [Dimmock, 2014], and the collection edited by Whittle is the collective stocktake on tenure [Whittle, 2013].

The second is *methodological*, and concerns whether England may be explained from inside England at all. Anievas and Nişancıoğlu's charge is that Brenner and Wood are internalist, and thereby Eurocentric [Anievas and Nişancıoğlu, 2015]; Heller presses Anglocentrism and a case for French capitalism [Heller, 2011]; Banaji rejects the rigid mode-of-production boundary the argument needs [Banaji, 2010]; van Bavel's Low Countries arrived without the English tenurial structure at all [van Bavel, 2010]. The current Political Marxist answer is the programme set out by Lafrance and Post [Lafrance and Post, 2019], with Lafrance's France [Lafrance, 2019] the counterpart the model's own France regime (§4.6) should be read against rather than Brenner's 1976 sketch. Bois's original jibe, that this is *political* Marxism because the politics carries the explanation [Bois, 1978], is the ancestor of a line pressed since by Heller and by Davidson [Davidson, 2012]; RQ3 below is its direct test. Moore's complaint, that the account brackets nature [Moore, 2003, Moore, 2015], is the one this model is unusually well placed to answer rather than concede, since it has an ecological driver Brenner does not (§4.4).

The model is exposed on the second front in a way worth conceding at the outset: a closed, England-shaped lattice (§4.2) is methodological internalism implemented in code. The defence is not that the charge does not apply but that it applies more usefully here than to prose, since an assumption written as a boundary condition can be stated, varied and reported, where the same assumption carried as a habit of narration cannot. No question below tests it, and §6 returns to what that costs.

3 Research Questions

The questions below are sorted on one distinction, and it is not the obvious one. The obvious line is between novel questions and sanity checks; the useful line is between **claims about the theory** (§3.1) and **claims about the model** (§3.2). A negative result in the first class is a finding about Brenner and Wood; a negative result in the second is a defect to repair before any finding can be reported at all. Filing a question in the wrong class is how a modelling failure comes to be published as a historical result, and how a genuine result comes to be buried among diagnostics. Two items earlier drafts treated as robustness – enclosure's separability from rent conversion, and whether demographic pressure suffices – belong on the theory side, and appear there as RQ9 and RQ7.

Each question is stated as a claim, against a named rival, with what the model can settle between them. This is deliberate. A question of the form "does the model reproduce X" puts nothing at risk, because for most X both answers are compatible with the theory; the form used instead is "the account commits us to X, and here is whether X survives".

This set is deliberately over-complete, and will be pruned. Ten theory-facing questions is more than one paper can carry, and which of them are worth carrying cannot be settled before the model's dynamics are understood: a question that turns out to have a degenerate answer, or whose answer is fixed by a construction rather than by a mechanism, should not survive to the final text. §3.3 records the current view of which are candidates for the main argument and which for an appendix, and that view is expected to change as §5 fills in.

3.1 Claims about the theory

RQ1: Does the improvement dynamic require competitive tenancy, or only market exposure under secure property? Brenner's mechanism is a landlord–tenant symbiosis in which competitive tenancy is what compels reinvestment, displacing "the traditionally antagonistic relationship in which landlord 'squeezing' undermined tenant initiative" [Brenner, 1976, p.65]. Earlier drafts posed this as improvement being a consequence rather than a precondition of market exposure, which put nothing at risk, since the model's own construction makes $\iota_j^T(t)$ a function of exposure and the answer could hardly come out otherwise. The live rival is Allen's: that the productivity gains came from yeomen and owner-occupiers, that is from *secure property* rather than from competitive tenancy [Allen, 1992]. The model can adjudicate because Freehold (§4.6) carries the same improving disposition through a shadow rent while paying no rent at all, so the comparison is $\iota_j^T(t)$ and $k_j(t)$ for Leasehold against Freehold, on the same land, in the same run. Brenner's claim predicts Leasehold leads; Allen's predicts Freehold does. The event study around each parcel's own conversion is retained as the within-tenure half of the same question, with the occupant-continuity diagnostic (§4.12) reported alongside it, since a conversion that replaces the sitting tenant records a change of occupant rather than a change of disposition.

That comparison is only meaningful if the shadow rent is not doing the work by itself, and as specified it may be: a Freeholder feels the full competitive yardstick $M_j(t)$ while paying $\rho_j(t) = 0$, so out-accumulating a Leaseholder is closer to an accounting consequence than to a result. The switch `freehold_shadow_rent` therefore brackets the question rather than answering it – with the shadow rent on, Allen's outcome is favoured by construction; with it off, Brenner's is imposed by fiat, since Freeholders then face no competitive test at all. Neither arm settles RQ1; the interval between them is the result, and if that interval contains both answers the honest report is that the model as specified cannot separate them.

RQ2: Do Brenner's own mechanisms generate a spreading transition, or only a simultaneous one? Every mechanism Brenner names is uniform across England: inflation erodes fixed customary rents everywhere at the same rate (§4.4), the entry-fine loophole is available to every lord [Brenner, 1976, p.62], and the state–landlord alliance is a national condition [Brenner, 1976, p.71-72]. Fiscal deficits therefore accrue to all landlords roughly together, and nothing in the stated account explains why the transition should begin anywhere in particular. Yet a transition occurring everywhere at once is neither the historical pattern nor what the theory is usually taken to describe. The test is the distance-dependent lag of §4.17(iii) under ablation of the two transmission channels of §4.7: if removing both yields simultaneity, the account is *incomplete as stated* and requires an inter-estate mechanism Brenner never specifies. That is a constructive result rather than a refutation – it identifies what the theory needs and does not have – and it is the kind of thing only a formalisation can establish, since prose can leave the question unasked.

There are two channels rather than three because tenant mobility, which an earlier draft treated as a third, is not one: the sources place the fight over mobility before the period as a settled precondition, and the version in which lords bid against one another to retain mobile tenants is Postan's, which Brenner reproduces only to reject (§4.7). Competitive allocation (§4.8) replaced it, and governs who ends up holding land rather than how conversion travels; it is therefore reported against concentration, not ablated as a spread channel.

RQ3: Is the state–landlord alliance necessary, sufficient, or merely permissive? Bois named this school *political* Marxism as an accusation: that the politics carries the explanation while class conflict decorates it [Bois, 1978], a charge pressed since by Heller [Heller, 2011] and Davidson [Davidson, 2012]. The model puts it to a direct test, because θ (§4.5) and the class-conflict mechanism are separable. A single England/France flip cannot answer it – that shows only that θ matters, which nobody disputes. What is required is the 2×2: England and France values of θ, each crossed with the class-conflict mechanism enabled and disabled, the latter operationalised as $\alpha_1 = 0$, so that a lord's fiscal pressure no longer drives conversion, together with $\varphi = 0$, removing the squeeze against which conversion has to compete (§4.9). If the England outcome survives with class conflict disabled, the politics is sufficient on its own and Bois was right. If θ merely widens or narrows a transition that class conflict drives either way, it is permissive, and the charge fails. The France arm should be read against Lafrance's account [Lafrance, 2019], in which France is not simply not-England but has a peasant-proprietor dynamic of its own – which is what the Freehold-diversion pathway of §4.6 actually models.

RQ4: How secure could customary tenure have been and still have given way? This is where the empirical front becomes tractable rather than merely threatening. Whittle finds custom more secure and the peasant land market more active than Brenner's account allows [Whittle, 2000]; Bailey finds its decline driven by law rather than by struggle [Bailey, 2014]; Dimmock disputes both [Dimmock, 2014]. What none of them can do is quantify how much security is *compatible* with the outcome, because that is a counterfactual and the evidence is a single realised history. The model can: sweeping the hazard of reopening terms ϖ, the arbitrary-fine rate φ and institutional resistance θ locates the frontier beyond which the transition fails to complete within the run. The result is a bound rather than a verdict – custom could have been this secure and England would still have got there, but no more secure than this – and it is the clearest single argument for using a model at all, since it answers a question the archive cannot be asked.

RQ5: Is dispossession necessary to the transition, or only sufficient? Whittle's stronger claim is that agrarian capitalism developed in Norfolk without the wholesale dispossession Brenner's narrative foregrounds [Whittle, 2000]. The model separates the routes by which a parcel changes hands (§4.12): succession vacancies, at which an heir is displaced only by an entry fine, are distinct from crisis vacancies, at which no claimant remains to contest the holding. Damping the crisis channel – $\eta_1 = \eta_2 = 0$, with eviction effectively suspended – leaves conversion to run through succession and fine escalation alone. If Leasehold dominance, farm concentration and a wage-labour pool still emerge, dispossession is sufficient but not necessary, which reconciles Brenner's logic with Whittle's evidence instead of forcing a choice between them. If they do not, Brenner's emphasis is vindicated on his own terms.

RQ6: Is ecology doing explanatory work the theory attributes to class structure? The model contains an exogenous ecological driver (§4.4) and a real spatial distribution of land quality (§4.2), neither of which Brenner has. This is a liability and an opportunity at once. Moore's charge is precisely that the account brackets nature [Moore, 2003, Moore, 2015], so if the timing and location of the transition here are set by soil rather than by class position, the model has demonstrated Moore's point using Brenner's own mechanisms. The ablation is `uniform_fertility`, which flattens the ALC-derived regional structure to its national mean while retaining parcel-level jitter, so that land quality loses its spatial *pattern* without losing its heterogeneity. Because the distance regressor still measures from the same county, a lag surviving the ablation is by construction not an ecological one. To our knowledge this comparison has not been made, and it is the most original question in the set.

RQ7: Does demographic pressure suffice, with class structure removed? Brenner's 1976 intervention was written against demographic determinism, and the model can re-stage that founding argument with an instrument neither side had. The population rule is switchable between a closed population, single-body vital rates and the full household demography of §4.13: running the transition with population fixed asks whether it proceeds without demographic pressure at all, and running it with births endogenous while the class mechanism is damped as in RQ3 asks whether demography alone can carry it. Earlier drafts filed this as robustness, which was a category error – "does the transition still occur with population held fixed" is not a sanity check on the model, it is the central claim of the position Brenner was answering.

RQ8: Does landlord–tenant symbiosis have a parameter range, and where are its edges? Brenner distinguishes English symbiosis, in which lord and tenant both accumulate, from continental squeezing, in which extraction undermines tenant initiative [Brenner, 1976, p.48, p.65], but gives the distinction no boundary. The extraction share θ_{rent} (§4.9) makes it quantitative: there should be a band within which both classes accumulate, and outside which the lord destroys the tenant's capacity to improve and with it his own rent roll. Sweeping θ_{rent} locates that band, or fails to find one – and a symbiosis obtaining at every extraction rate is a description rather than a mechanism.

RQ9: Is enclosure a driver of proletarianisation, or a consequence of it? Wood treats the extinguishing of common right as a process distinct from rent conversion though legitimated by the same improvement ethic [Wood, 2002, p.108-115], and Neeson's account makes the loss of common right substantial and separable [Neeson, 1993]. Shaw-Taylor pushes back, dating proletarianisation later and reducing enclosure's share of it [Shaw-Taylor, 2012]. In the model $\Xi(t)$ is driven by the diffusing ideology, itself driven by realised conversions (§4.11), so enclosure is *downstream* by construction; what remains open is whether the landless pool would exist without it, which $\nu_1 = \nu_2 = \nu_3 = 0$ answers directly. This too was filed as robustness and does not belong there: whether the channels are separable is exactly what Wood asserts and Shaw-Taylor denies.

3.2 Claims about the model

The items below are not sources of novelty and are not evidence about the theory. They ask whether the model behaves well enough for anything above to be reported at all, in the way a calibration check would. A negative result here is a defect to repair, not a finding to publish.

- *Sufficiency of the aggregate transition.* From a Customary-dominated initial condition, does the England-regime run move toward Leasehold dominance at the population level at all (§4.17(i))? This is a precondition for RQ1 and RQ2 to be meaningful, not itself a claim under test.
- *Attribution of concentration.* Does the Gini coefficient of tenant holding size rise over the run, and is the rise attributable to engrossment (§4.10) specifically rather than to ordinary population turnover (§4.17(ii))? Concentration produced by turnover alone would leave RQ1's Leasehold/Freehold comparison uninterpretable, since holding size enters output.
- *Macro plausibility.* Does the landless pool show both continued agricultural hiring and a growing cumulative exit to industry, does the wage series diverge from rents in the direction the record suggests [Kerridge, 1953, Clark, 2002], and is aggregate output's shape consistent with the secondary series of §4.17 (targets (iv) and (vi))? Qualitative throughout: §4.16 disclaims calibration, so a mismatch in level is not a failure and a mismatch in direction is.
- *Population accounting.* Is the population closed in the sense §4.12 claims – no unmodelled reservoir of prospective tenants, and no silent depopulation from an absorbing urban state – so that every share is a share of a determinate total? Two diagnostics, both of which failed under the earlier closed population: whether the share of parcels no household can take up stays near zero, and whether cross-seed variation in tenure outcomes is independent of cross-seed variation in population. This is the check that keeps RQ7 a question rather than an artifact, and it is filed here rather than with RQ7 precisely because the two are easy to confuse: the accounting must hold under every population rule before any comparison between them means anything.
- *Independence from the demographic regime.* Distinct again from RQ7: not "does demography drive the transition" but "does the answer to any other question depend on which demographic rule is in force". A result holding only under one population rule is not a result about property relations, and belongs in neither class until it is understood.

3.3 Which questions are expected to survive

Recorded here so that the pruning is visible rather than quiet. On the current view the main argument is carried by RQ1–RQ4 together with RQ6: one question per live front – the improvement mechanism against Allen, the completeness of the account against itself, the political charge against Bois and Heller, the empirical challenge against Whittle and Bailey, and the ecological one against Moore. RQ7 sits between the two groups and should be promoted if the fixed-population arm produces a transition and demoted if it does not, since a null there speaks to the model's calibration rather than to Postan.

RQ5, RQ8 and RQ9 are the appendix candidates. Each refines a question already in the main set rather than opening a front: RQ5 varies the channel through which RQ4's custom gives way, RQ8 is the parameter-range version of RQ1's mechanism, and RQ9 separates a channel whose downstream position is already fixed by construction. The full four-way ablation table behind RQ2 likewise belongs in an appendix, with only the headline comparison in the text. The allocation is provisional in both directions: a question returning a sharp and unexpected answer earns its way into the main argument whichever group it starts in, and any question whose answer proves to be fixed by a construction rather than by a mechanism should be dropped outright rather than relegated.

4 Model Design

The model is designed as a test of the internal consistency of the Brenner–Wood account, rather than as a stylised illustration of it. The question it asks is whether the class-conflict mechanisms Brenner and Wood identify are jointly sufficient to generate, from a small and spatially localised seed, the macroscopic outcomes the theory is invoked to explain: the collapse of customary tenure, the concentration of landholding, the emergence of the landlord/capitalist-tenant/wage-labourer triad [Brenner, 1976, p.63], and the spread of an "improvement" ethic that legitimises enclosure [Wood, 2002, p.108-109]. Every modelling choice below reflects a specific claim in the primary literature, cited with a page reference, so that the model's assumptions remain traceable to – and falsifiable against – the theory it is testing.

4.1 Agents

Landlord $i \in \{1, \ldots, N_L\}$. Each landlord holds an estate of tenancies $\mathcal{T}_i$, a wealth stock $W_i(t)$, and a fixed consumption requirement C_i tied to the reproduction of aristocratic class position rather than to profit-maximisation as such [Brenner, 1976, p.31]. Landlords do not choose to become capitalists; they act defensively, to avoid losing their own class position, and any systemic shift toward market dependence is a by-product of that defensive action, both lords and peasants "involuntarily setting in train a capitalist dynamic while acting in class conflict with each other to reproduce themselves as they were" [Wood, 2002, p.52]. Each landlord also has an awareness radius over neighbouring estates' rents and outcomes [Wood, 2002, p.100-101], and an exposure to "improvement" ideology $\iota_i(t)$.

Tenant household j. Rather than instantiating landlord, capitalist-tenant and wage-labourer as three separate agent classes from the outset – which would hard-code the very triad the theory is meant to explain – a tenant is a single agent type carrying a tenure state $\sigma_j(t) \in \{\text{Customary}, \text{Leasehold}, \text{Freehold}, \text{Landless}\}$ that evolves endogenously, and a holding $\mathcal{P}_j(t)$ – a set of land parcels, since holdings themselves grow and shrink through engrossment (§4.10) rather than being fixed at one parcel per tenant. This reflects Brenner and Wood's own emphasis that the triad is an *outcome* of class conflict, not a precondition of it [Brenner, 1976, p.63]; Freehold is included as the historical alternative outcome of the same conflict that prevailed in France rather than England (§4.6).

The agent is a *household* rather than a person, and carries a size $n_j(t) \in \{1, 2, \ldots\}$ in adult labour equivalents. It supplies $n_j(t)\bar{\ell}$ labour to its own holding and owes $n_j(t)\, c^{\text{head}}$ in subsistence, so a member is simultaneously a pair of hands and a mouth; $n_j(t)$ then evolves through the fertility, mortality and partition rules of §4.13. Two things follow that a single-body agent cannot represent. First, the peasant family farm's characteristic response to pressure – absorbing more labour on the same land rather than buying it in – becomes available to the model, and because only *hired* labour is charged at $\omega(t)$ in (13) while family labour is not, the family farm has a genuine cost advantage over an equally sized holding worked by wage labour [Chayanov, 1966]. Second, and more consequentially for the argument, it makes proletarianisation something a household can *produce* – the member for whom the holding has no profitable use, historically the non-inheriting son – and not merely something eviction does to it.

Land parcel k, held by a tenant, with a fertility/productivity state $\phi_k(t)$ subject to an exogenous ecological-demographic process (§4.4). Land is not a decision-maker, but its local, heterogeneous condition is what gives the model's initial "seed" of competition a non-arbitrary spatial location, rather than requiring the modeller to plant it by hand.

4.2 Spatial structure: an England-shaped lattice

Awareness radius, parcel adjacency, and the "distance-dependent lag" validation test (§4.17) all presuppose an explicit space. Rather than an abstract torus, the lattice is laid over the real outline of England and its heterogeneity is drawn from a real, if necessarily modern, proxy for regional land quality – so that the model's "small, spatially localised seed" (§4) is a location the data itself supplies, not one the modeller plants by hand. England, rather than Great Britain, is the deliberate extent: θ, the scenario parameter that does the real explanatory work in §4.5, stands for a specifically English alignment of Crown and gentry that Brenner and Wood both treat as sui generis to England [Brenner, 1976, Wood, 2002, p.68-75], and Scotland in particular had its own distinct legal and tenurial history through this period; including it would blur the very comparison the model is built to make, of which the France regime (§4.5) is already the intended counterfactual.

Construction (a one-off preprocessing step, not a runtime dependency). Land parcels k occupy cells of a lattice laid over England's bounding box at resolution L: the longer axis of the box is divided into L cells and the shorter axis takes as many square cells of that same size as it holds, so that a lattice distance is proportional to a real distance along both axes. At $L = 155$ this is a 155×136 grid, of which 7,417 cells survive once those falling outside the country's boundary [Office for National Statistics, 2022] are excluded outright – no wraparound: unlike the earlier torus, a real coastline and the Scottish and Welsh borders are historically meaningful edges, not artifacts to be engineered away. Each surviving cell is assigned, once, to whichever of England's 39 historic counties its centroid falls within, and inherits that county's fertility baseline $\bar{\phi}^{\text{county}}$, computed by area-weighting Natural England's Agricultural Land Classification (ALC) grades [Natural England, 2021] within the county (Grade 1, "excellent", mapped to the highest score, Grade 5, "very poor", to the lowest).[2] This lookup of ≈40 numbers, one per historic county, is all the running simulation ever needs – the GIS work of intersecting ALC polygons with county boundaries happens once, offline, in producing it. Each parcel then draws

$$\bar{\phi}_k \sim \mathcal{U}\big(\bar{\phi}^{\text{county}(k)} - \zeta,\ \bar{\phi}^{\text{county}(k)} + \zeta\big), \tag{1}$$

retaining fine-grained local heterogeneity within a county (so engrossment and turnover still vary parcel-to-parcel, §4.10) while centring it on a real regional value: modern ALC grades reflect a soil-and-climate endowment that changes on a geological, not a century, timescale, so treating it as invariant over the model's $\sim$150-year window is a stated proxy assumption, distinct from the fast-moving, endogenous depletion and regrowth (3) already gives each parcel. The model's low-fertility "seed" is then simply wherever this construction happens to put the worst county – historically the upland, wood-pasture and moorland counties rather than the arable South and East – with no separate seed-radius parameter required.

[2] Two notes on the county set, both recording corrections. The ONS Ancient Counties layer [Office for National Statistics, 2022] lists the County of London separately from Middlesex; it is excluded, because it is a city rather than an agrarian county and Natural England returns no graded agricultural land inside it at all. Retaining it meant it fell through to the national mean baseline, crediting the capital with average farmland; excluding it leaves its cells to the nearest-county assignment, which returns them to the surrounding counties whose land they were, and brings the total to exactly thirty-nine. Separately, an earlier version of the artifact held only thirty-seven counties: the filter selecting English counties out of the England-and-Wales layer tested a mean of boundary *vertices* rather than an area-weighted centroid. A vertex mean weights a boundary by how finely it happens to be sampled, so Kent, Cheshire and Somersetshire – each with a long, heavily-vertexed estuarine coast, respectively the Thames and Medway, the Dee and Mersey, and the Severn – had their apparent centre dragged out to sea and were silently dropped. The criterion is now the share of a county's sampled interior falling inside England, which separates cleanly: every English county scores above 0.98 and every Welsh one below 0.01.

N_L landlord seed points are scattered uniformly at random within *each* historic county (uniform rather than weighted toward more heavily manorialised regions, since weighting would require a second historical dataset – population or manor density circa 1600 – and would start to blur the model from a theory-test with a realistic backdrop into a historical reconstruction, which §4 explicitly disclaims). Estates are then formed by a Voronoi tessellation around these seeds *within each county*: parcel k belongs to landlord i's estate $\mathcal{T}_i(0)$ if i's seed is k's nearest seed among those falling in k's own county. Seeding per county rather than nationally is what makes the county a genuine upper tier rather than a label: no estate straddles a county boundary, so each estate inherits exactly one fertility baseline $\bar{\phi}^{\text{county}}$, and the ecological seed is an estate-level condition rather than one cutting across estates. This gives each landlord a contiguous but irregularly shaped estate without requiring the modeller to hand-draw manor boundaries – a representational choice, not a historical claim, in the same sense that the logistic form of (3) is a standard functional choice rather than one drawn from Brenner or Wood.

Two distinct neighbour relations are read off the same lattice, since Brenner and Wood's own language keeps them at different levels: engrossment operates "within the individual tenancy relation" (§4.11), i.e. between adjacent *parcels* under a single landlord, whereas the spread channels of §4.7 operate between *estates*.

- **Parcel adjacency** $\mathcal{N}(k)$ (§4.10): the Moore (8-connected) lattice neighbours of k that belong to the same landlord's estate as k.
- **Landlord awareness radius** $\mathcal{N}_i$ (§4.5, §4.7): all landlords $i' \neq i$ whose seed point lies within Euclidean distance r of i's seed point.

The validation test of §4.17(iii) regresses each parcel's first-conversion time on its lattice distance from the centre of the lowest-fertility county identified above.

4.3 Causal structure

Figure 1 summarises how the mechanisms below connect; citations for each node are given in the corresponding subsection of the text. Two features are worth flagging before the equations. First, there are two, not one, channels by which a single landlord's break from custom propagates to others – observed neighbour outcomes, and the slower diffusion of a legitimating ideology – implemented as separable switches so that their relative importance can be tested rather than assumed. (The diagram retains a tenant-mobility node, which §4.7 withdraws; competitive allocation, §4.8, takes its place and is omitted here for legibility.) Second, the diagram makes explicit that *why* a landlord can break custom at all is treated as a structural, largely exogenous condition of the English case (the state–landlord alliance θ), not something the model derives from local bargaining – following Brenner's own comparative argument (§4.5).

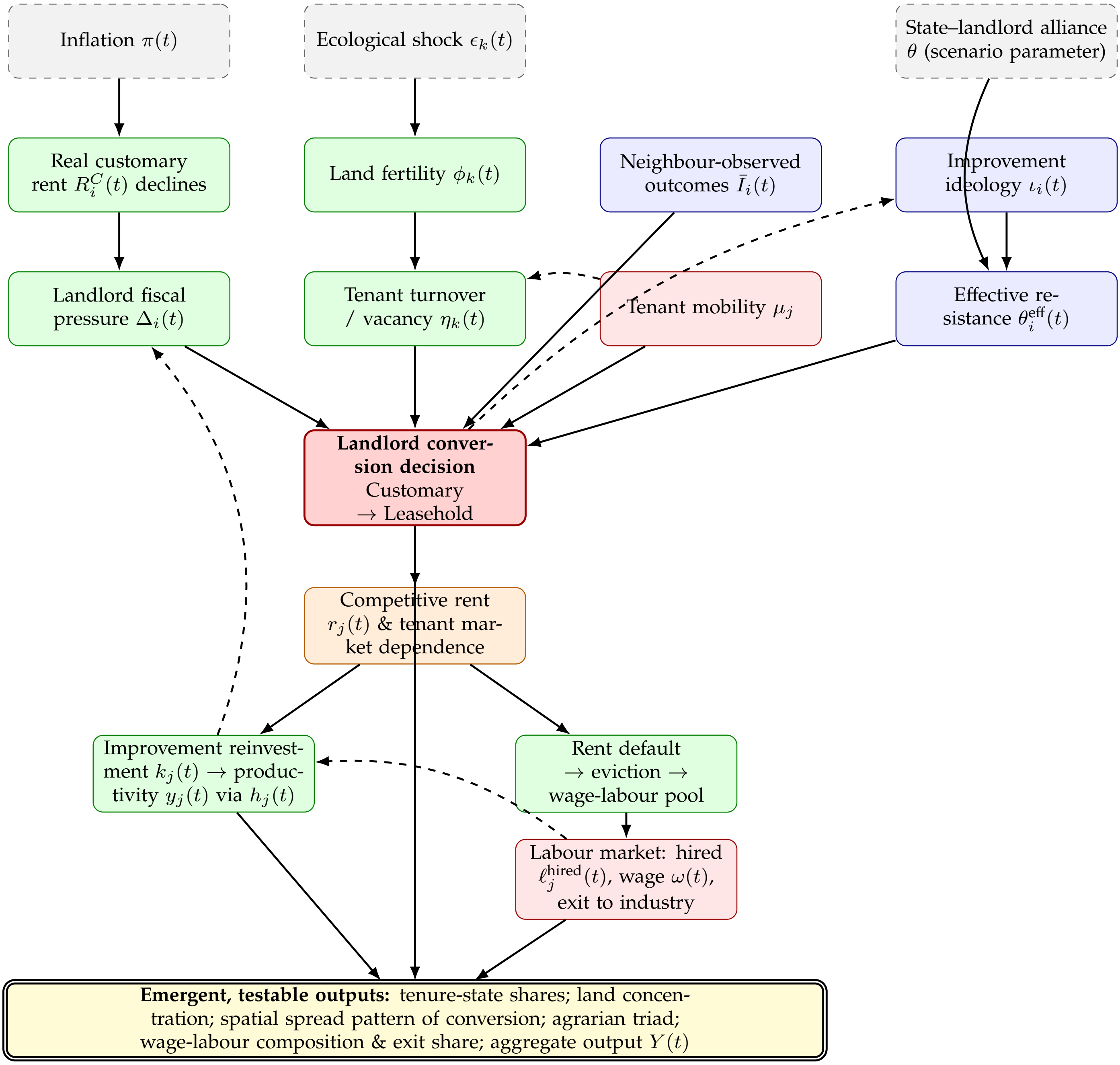

Figure 1: Causal structure of the model. Dashed boxes are exogenous/scenario parameters; solid green boxes are endogenous processes; blue boxes are the three spread channels of §4.7; the red box is the landlord's discrete decision; the double yellow box lists the emergent quantities used as validation targets in §4.17. Dashed arrows denote feedback loops (mobility creating a further vacancy; a conversion seeding the ideology that lowers resistance to the next conversion; tenant improvement reducing the landlord's own fiscal pressure; hired labour from the wage-labour pool raising the hiring tenant's own productivity, §4.14). Freehold entrenchment (§4.6) and enclosure of the commons (§4.11) are omitted from the diagram for legibility and are given only in equations.

4.4 Exogenous drivers: why feudalism cannot reproduce itself

Two exogenous processes represent the standing pressure that makes the customary order unsustainable on its own terms, independent of any competitive dynamic.

Inflation. Customary rent $r_j^C(0)$ is fixed in nominal terms at the point a tenancy is granted and is not renegotiated while a tenant remains in the Customary state: "rent *per se* (*redditus*) was fixed by custom, and subject to declining long-term value in the face of inflation" [Brenner, 1976, p.61]. With an exogenous price index $\Pi(t) = \prod_{s=1}^{t}(1 + \pi(s))$, the real value of landlord i's customary income is

$$R_i^C(t) = \sum_{j \in \mathcal{T}_i^C(t)} \sum_{k \in \mathcal{P}_j(t)} \frac{r_k^C(0)}{\Pi(t)}, \tag{2}$$

which decays monotonically even though nothing in the local land market has changed.

Ecological-demographic cycle. Land fertility is modelled as a renewable stock subject to logistic regrowth toward a parcel-specific carrying capacity $\bar{\phi}_k$ – the standard formalism for a resource that grows fastest at intermediate stock levels and saturates as it approaches its ecological ceiling [Verhulst, 1838], here combined with an extraction/shock term exactly as in the bioeconomic model of a harvested renewable resource [Schaefer, 1954] – punctuated by exogenous shocks $\epsilon_k(t)$:

$$\phi_k(t+1) = \phi_k(t) + \rho\, \phi_k(t)\Big(1 - \frac{\phi_k(t)}{\bar{\phi}_k}\Big) - \epsilon_k(t). \tag{3}$$

Neither Brenner nor Wood specifies a functional form for soil-fertility dynamics – nor, on inspection, does Jason W. Moore's account of feudalism's own ecological contradictions [Moore, 2003], which is historical-theoretical rather than mathematical. Their evidence is qualitative, concerning the exhaustion of arable land under insufficient manuring and the extension of cultivation onto marginal soils, and it is this qualitative claim, not a specific equation, that (3) is chosen to instantiate: the lord's surplus extraction "tended to confiscate not merely the peasant's income above subsistence...but at the same time threatened the funds necessary to refurbish...productivity" [Brenner, 1976, p.48], producing a "vicious cycle of the destruction of the peasants' means of support" in which "the crisis of productivity led to demographic crisis" [Brenner, 1976, p.49-50]. The heterogeneity of $\bar{\phi}_k$ across parcels is what gives the model's initial shock a location: some estates are ecologically buffered and some are not, mirroring this uneven pattern of subsistence crisis. A tenancy's turnover hazard rises when output falls below a subsistence threshold $\bar{y}$,

$$\eta_j^{\text{crisis}}(t) = \eta_1 \max\big(0,\ \bar{y} - y_j(t)\big), \qquad \eta_j^{\text{succ}}(t) = \eta_0, \tag{4}$$

drawn as two separate events on the *holding* rather than as one combined hazard on the parcel, because the two resolve differently at a vacancy (§4.12): a succession leaves a customary claimant, a crisis does not. The unit being the household's whole holding, $y_j(t)$ is its total output, so a holding that has engrossed its way well clear of $\bar{y}$ carries no crisis hazard from this term at all. Rent-default eviction, $d_j(t) = \tau$, takes precedence over both, and a crisis over a succession. So vacancies – through death, flight, or failure to hold the tenancy – cluster where ecological conditions are worst, exactly as Brenner describes for the fourteenth–fifteenth century demographic collapse: "the demographic collapse...left vacant many former customary peasant holdings. It appears often to have been possible for the landlord simply to appropriate these and add them to his demesne" [Brenner, 1976, p.61].

4.5 The landlord's conversion decision

At each vacancy, landlord i chooses between renewing the tenancy on customary terms or converting it to a leasehold let at a competitive rent. Define fiscal pressure as the gap between the landlord's consumption requirement and realised receipts, expressed as a share of that requirement,

$$\Delta_i(t) = \frac{C_i - R_i^C(t) - R_i^L(t) - F_i(t)}{C_i}, \tag{5}$$

where $F_i(t)$ is the arbitrary-fine income of §4.9. Two features of this deserve comment, because the obvious alternative is misleading. First, pressure enters as a *relative* shortfall: an absolute gap would mix a per-landlord requirement against rent summed over the estate, so a small estate would register far more pressure than a large one for purely arithmetic reasons, and estate size rather than any mechanism in the theory would determine who converted first. Second, C_i is itself set as a multiple of the estate's own initial rent roll rather than drawn independently of it. Drawn independently, every lord begins in heavy deficit and conversion is already likely everywhere in the first period, which leaves no interval during which custom is sustainable and therefore nothing for a localised shock to break – the dynamic §4.4 is built to produce. Scaling the requirement to the estate makes $\Delta_i(0) \approx 0$, so that fiscal pressure is genuinely *created* by the inflationary erosion of a fixed nominal rent.

The conversion probability is

$$p_i^{\text{convert}}(t) = \sigma\Big(\alpha_0 + \alpha_1 \Delta_i(t) + \alpha_2 \bar{I}_i(t) - \alpha_3\, \theta_i^{\text{eff}}(t)\Big), \tag{6}$$

with α_0 and α_1 calibrated so that complete erosion of customary income is *necessary but not sufficient*: a lord with no converted neighbours still converts only rarely, and it takes the neighbour term $\alpha_2 \bar{I}_i(t)$ to carry the decision. This is not a free choice of parameters but a restatement of the paper's own claim in §4.7 – that uniform inflationary pressure cannot by itself generate a spreading rather than a simultaneous pattern. Coefficients that let $\alpha_1 \Delta_i(t)$ saturate the logit on its own would assume away the question RQ2 asks. with $\sigma(\cdot)$ the logistic function, $\bar{I}_i(t)$ the neighbour-observation term (§4.7), and $\theta_i^{\text{eff}}(t)$ the effective strength of institutional resistance a tenant can muster against conversion (§4.7). This is also the point at which landlords historically had a second, non-confrontational lever besides outright conversion: raising the entry fine charged at conveyance, since "there often remained one crucial loophole open to the landlord...He could insist on the right to charge fines...whenever peasant land was conveyed...in the end entry fines appear to have provided the landlords with the lever they needed to dispose of customary peasant tenants, for in the long run fines could be substituted for competitive commercial rents" [Brenner, 1976, p.62]; the model treats a fine escalation as a special case of the same conversion decision, since its economic effect – a rent that tracks the market rather than custom – is identical.

Conversion in place. That last point has a consequence the vacancy framing obscures. If conversion is only ever reached *at* a vacancy, and the incoming occupant is drawn from the landless pool (§4.12), then scarcely any tenant is present both before and after the conversion of their own holding – and the central question of §3, whether market exposure changes a tenant's disposition, becomes unaskable, since what the data would record is a change of occupant. But the entry fine was levied on the sitting tenant, and it is precisely by that route that "in the long run fines could be substituted for competitive commercial rents" [Brenner, 1976, p.62]. The model therefore also lets a lord reopen terms mid-tenancy: with per-period hazard ϖ the decision above is applied to a sitting customary tenant, who either meets the fine $\chi\, \hat{r}_i(t) |\mathcal{P}_j(t)|$ and continues as a leaseholder, or cannot and is dispossessed, the holding passing to the market. Conversion by re-letting and conversion by fine escalation are the same decision reached by two routes, and only the second leaves a tenant to observe.

Crucially, θ is not derived endogenously from a submodel of peasant revolt. Brenner's own comparative argument treats the presence or absence of an independent political ally for the peasantry as the decisive, largely exogenous condition distinguishing England from France: in France "the monarchy...had an interest in limiting the landlords' rents...and thus in intervening against the landlords to end peasant unfreedom and to establish and secure peasant property" [Brenner, 1976, p.69], whereas in England the same centralising process ran through, not against, the landlord class – "important sections of the English nobility and gentry were willing to support the monarchy's centralizing political battle against the disruptive activities of the magnate-warlords...but it was precisely these same landlord elements who were most concerned to undermine peasant property" [Brenner, 1976, p.71-72]. Wood's account of the English state makes the same point institutionally: the aristocracy was "demilitarized before any other aristocracy in Europe...part of the increasingly centralized state...[but] did not possess autonomous 'extra-economic' powers...to the same degree as their continental counterparts" [Wood, 2002, p.99], so that landlords "depended less on their ability to squeeze more rents out of their tenants by direct, coercive means than on their tenants' success in competitive production" [Wood, 2002, p.100]. Accordingly θ is set as a scenario-level parameter – an "England" regime and a counterfactual "France" regime, used for validation in §4.17 – rather than as an emergent variable, with individual conversion attempts still subject to idiosyncratic stochastic resistance through the logistic error term, representing the real but ultimately unsuccessful local resistance of the sort seen in Ket's Rebellion of 1549: "if successful, the peasant revolts of the sixteenth century...might have ‘clipped the wings of rural capitalism’...But they did not succeed” [Brenner, 1976, p.62-63].

4.6 Freehold tenure and the closure of an alternative path

The conversion decision above is not the only possible outcome of a customary tenant’s struggle against a landlord under fiscal pressure. In the mid-fifteenth century, before landlords found the entry-fine loophole, the outcome was genuinely open: “peasant tenants at this time were striving hard for full and essentially freehold control over their customary tenements, and were not far from achieving it” [Brenner, 1976, p.61]. Freehold is therefore included as a fourth tenure state, not merely as a residual category, because the theory’s comparative logic requires a place for the path England did *not* take.

Initial condition, not emergent outcome. Consistent with Wood’s account, the initial share of Freehold tenants is set low and is not itself a target the England-regime run is expected to explain: “the concentration of English landholding meant that an unusually large proportion of land was worked not by peasant-proprietors but by tenants...This was true even before the waves of dispossession...associated with ‘enclosure”’ [Wood, 2002, p.99-100], in contrast to France, which “remained a country of peasant proprietors” while “land in England was concentrated in far fewer hands” [Wood, 2002, p.133].

Freeholders still compete. Owning land outright does not exempt a tenant from the market imperative, since the compulsion runs through the goods market as much as the rental market: “yeomen were typically the very kind of capitalist tenants who were subject to competitive pressures, and even owner-occupiers would be subject to those pressures once the competitive productivity of agrarian capitalism set the terms of economic survival” [Wood, 2002, p.54]. A Freehold tenant’s wealth therefore evolves by the same unified rule as any other tenure state ((13), §4.9, with $\rho_j(t) = 0$), using the same production function (18) and the same competitively-driven improving disposition $\iota_j^T(t)$ and reinvestment rate $\bar{s}\,\iota_j^T(t)$ (§4.9, (15)), with the estate’s leasehold benchmark $\hat{r}_i(t)$ standing in as j’s shadow rent for the purposes of $\hat{M}_j(t)$ even though none is actually paid; a Freeholder who cannot sustain c_j for τ periods sells up and enters the Landless state, while one who prospers can absorb a neighbouring vacant parcel through the same engrossment competition available to capitalist tenants (§4.10) – “the same would increasingly be true even of landowners working their own land. In this competitive environment, productive farmers prospered and their holdings were likely to grow, while less competitive producers went to the wall and joined the propertyless classes” [Wood, 2002, p.103].

A rare, closing pathway. At a vacancy where the sitting customary tenant’s resistance succeeds (§4.5), there is a further, small probability that the outcome entrenches as Freehold rather than reverting to Customary,

$$p_j^{\text{Freehold}}(t) = \sigma\big(\lambda_0 + \lambda_1\,\theta - \lambda_2\,\mathbb{1}[t \geq t^*]\big), \tag{7}$$

increasing in the same state–landlord alliance parameter θ used throughout, and falling discretely once the entry-fine loophole fully closes at the same t^* that marks the end of state resistance to enclosure (§4.11) – formalising the closing-off already asserted qualitatively below. In the England regime (θ low) this is close to zero even before t^*; in the France regime (θ high) this pathway dominates, giving the model’s England/France comparison (§4.17) a mechanism rather than only a suppressed probability – the same underlying pressure resolves to the Leasehold triad in one regime and to entrenched Freehold smallholding in the other, exactly the divergence Brenner draws out at length in his own comparison of the two countries’ seventeenth-century agrarian structures [Brenner, 1976, p.68-75].

4.7 Spread of the market imperative

Given uniform inflationary pressure, fiscal deficits accrue to all landlords roughly together; what produces a spreading, rather than simultaneous, pattern of conversion is that converting is a costly institutional break, so landlords are heterogeneously reluctant to be first. Two channels transmit a successful defection to its neighbours: what a lord can observe of a neighbour’s results, and the slower diffusion of a legitimating ethic.

Why not tenant mobility. An earlier version of this model treated tenant mobility as a third transmission channel, on the reasoning that a tenant leaving for better terms elsewhere forces the losing lord’s hand.

That has to be withdrawn, for two reasons. The first is chronological: the fight over mobility is settled before the period modelled here begins. Controls on movement were constitutive of serfdom – their point was "preventing peasant mobility and thus a 'free market in tenants"' [Brenner, 1976, p.44] – but "by the mid-fifteenth century, through flight and resistance, [the peasantry was able] to break definitively feudal controls over its mobility and to win full freedom" [Brenner, 1976, p.61]. Mobility is therefore a standing precondition of the model, not a variable within it. The second reason is more serious: in the sixteenth century, with population rising, the competitive pressure runs the other way. "Competition for land induces the peasantry to accept a serious degradation of their personal/tenurial status in order to hold on to their land" [Brenner, 1976, p.38], and lords, far from bidding to retain tenants, "could always get replacements, quite often indeed on better terms" [Brenner, 1976, p.44]. Lords competing to keep mobile tenants is Postan's mechanism, which Brenner reproduces only to reject (§1.2); building it in would smuggle the demographic account into a model meant to test the class-structural one.

What the sources do describe at this point is not migration but *allocation*, and it is given its own subsection below (§4.8).

Observed outcomes. Landlords within i's awareness radius $\mathcal{N}_i$ update on the realised wealth gain of neighbours in proportion to how much of those neighbours' own estates have already converted – a landlord's estate converts tenancy-by-tenancy, not all at once (§4.12), so the signal is a continuous Leasehold share $s_{i'}^L(t) = |\mathcal{T}_{i'}^L(t)|/|\mathcal{T}_{i'}(t)|$ rather than a binary "has converted" flag,

$$\bar{I}_i(t) = \frac{1}{|\mathcal{N}_i|} \sum_{i' \in \mathcal{N}_i} s_{i'}^L(t) \cdot \frac{W_{i'}(t)}{W_{i'}(0)}, \tag{8}$$

with $t = 0$ the simulation's start, at which $W_{i'}(0) > 0$ by construction (§4.16), and $|\mathcal{T}_{i'}(t)|$ in $s_{i'}^L(t)$ counting only currently active, separately-let tenancy slots on estate i' – a parcel absorbed by engrossment (§4.10) has ceased to be its own tenancy and drops out of both numerator and denominator. This directly reflects the historical practice Wood describes of landlords' surveyors explicitly pricing the gap between fixed and competitive rent – "we can watch the development of a new mentality by observing the landlord's surveyor as he computes the rental value of land...against the actual rents being paid by customary tenants...[calculating] 'the annual value beyond rent' or 'value above the oulde [sic] rent"' [Wood, 2002, p.101].

Ideological legitimation. Adoption of "improvement" as a legitimating ethic diffuses across landlords as a Bass-type contagion process, with the contagion term driven by neighbours' realised *conversions* $s_{i'}^L(t)$ rather than by their professed ideology,

$$\iota_i(t+1) = \iota_i(t) + \beta_1\big(1 - \iota_i(t)\big) + \beta_2\, \bar{s}_{\mathcal{N}_i}^L(t)\big(1 - \iota_i(t)\big), \tag{9}$$

and with β_1, the autonomous term, kept small. Letting ideology diffuse from ideology alone would make neither term refer to any actual break from custom, so ι_i would climb toward unity on a schedule of its own whether or not a single tenancy had converted – an exogenous clock rather than a transmission channel, and one whose ablation in §4.17 would test nothing. Figure 1 already draws the arrow from the conversion decision back into ideology; this is the equation that implements it. and lowers the effective force of institutional resistance faced at conversion,

$$\theta_i^{\text{eff}}(t) = \theta \cdot \big(1 - \xi_0 - \xi_1 \iota_i(t)\big). \tag{10}$$

This is the mechanism Wood documents in detail: enclosure and the extinguishing of customary right were pursued not only economically but legally and rhetorically, since "between the sixteenth and eighteenth centuries, there was growing pressure to extinguish customary rights that interfered with capitalist accumulation...traditional conceptions of property had to be replaced by new, capitalist conceptions of property" [Wood, 2002, p.108], with "reasons of 'improvement'...cited systematically as the basis of title to property and as grounds for extinguishing traditional rights" [Wood, 2002, p.115], of which Locke's labour theory of property was the paradigm articulation [Wood, 2002, p.108-112].

Both channels are implemented as independent switches so that the paper's Results section can ask which, if either, is necessary to reproduce a spatially spreading – rather than simultaneous – pattern of conversion: the model's central internal-validity test.

4.8 Competitive allocation: letting to the highest bidder

Withdrawing mobility does not remove competition between tenants; it relocates it. What Wood describes is a market in which lords choose among rival takers rather than tenants choosing among rival lords:

> "landlords...would increasingly seek tenants who could produce competitively and pay good rents by increasing their own productivity. This meant that success would breed success, and competitive farmers would have increasing access to even more land, while others lost access altogether...In a system of 'competitive rents', in which landlords, wherever possible, would effectively *lease land to the highest bidder, at whatever rent the market would bear*" [Wood, 2002, p.101].

This is an allocation rule, and the model implements it as one. When a market-exposed parcel k falls to be let (§4.12), the eligible takers bid what the holding is worth to them, which given (18) is a share of the output each could expect to raise on it:

$$b_j(k) = \theta_{\text{rent}}\, P(t)\cdot \hat{y}_j(k), \qquad \hat{y}_j(k) = \begin{cases} y_j(t)\,/\,|\mathcal{P}_j(t)| & j \text{ a sitting market-exposed neighbour,} \\ \phi_k(t)^{\mu}(k^{\text{trad}})^{\gamma}(n_j\bar{\ell})^{1-\mu-\gamma} & j \text{ from the Landless pool,} \end{cases} \tag{11}$$

struck in money, since a rent denominated in bushels would not be comparable with the income it is paid out of. A sitting neighbour therefore bids on *revealed* productivity – what they are actually getting per parcel already – while a landless entrant can only bid what an unimproved holding worked by its own family would yield. Household size enters the entrant's bid through its labour $n_j(t)\bar{\ell}$, so a larger landless household outbids a smaller one: the family-labour advantage, priced into the auction. Landless households differ only in wealth and in how many hands they bring, and both raise what they can bid, so it is enough to enter the best of each – the wealthiest and the largest. The parcel goes to the highest bidder who can also meet the entry fine $\chi\, b_j(k)$, and the winning bid becomes that parcel's rent: this is what "whatever rent the market would bear" means, and it makes the competitive rent an outcome of the auction rather than a parameter imposed on it. Adjacent tenants enter the auction only if the lord is willing to consolidate, which is the engrossment draw $q_i(t)$ of §4.10; the two are the same event seen from opposite sides, the lord's decision whether to break the holding up and the successful farmer's opportunity to absorb it.

The consequence is cumulative and one-directional, which is Wood's point. A tenant who has improved bids from a higher productivity, wins more land, and spreads the same capital over more parcels; a tenant who has not is outbid and eventually has nothing to bid with. Note that this is a mechanism of *concentration*, not of *transmission*: it governs who ends up farming, not how the break from custom travels between estates. It is therefore reported separately from the two spread channels above, and is not one of the switches §4.17 ablates when asking what produces contagion.

4.9 Tenant response: market dependence and improvement

Every occupied tenancy, whatever its tenure state, faces a state-dependent rent

$$\rho_j(t) = \begin{cases} \sum_{k\in\mathcal{P}_j(t)} r_k^C(0)\,/\,\Pi(t) & \sigma_j(t) = \text{Customary,} \\ \sum_{k\in\mathcal{P}_j(t)} r_k(t) & \sigma_j(t) = \text{Leasehold,} \\ 0 & \sigma_j(t) = \text{Freehold,} \end{cases} \tag{12}$$

the rent accruing *per parcel* rather than per tenancy, since a rate charged once per tenancy would let a tenant engross five parcels and pay what they paid for one – making concentration rent-free and over-incentivising the very outcome §4.10 is meant to explain. Each leasehold parcel carries its own rent $r_k(t)$, fixed at the bid that won it at the auction of §4.8 and reset only when the parcel is next let; where competitive allocation is switched off, every leasehold parcel on estate i instead carries the estate benchmark $\hat{r}_i(t)$, so that $\rho_j(t) = \hat{r}_i(t)\,|\mathcal{P}_j(t)|$.

Wealth then evolves by a single rule regardless of state,

$$w_j(t+1) = w_j(t) + R_j(t) - \rho_j(t) - n_j(t)\,c^{\text{head}} - \omega(t)\,\ell_j^{\text{hired}}(t) - F_j(t), \tag{13}$$

where the credited quantity is revenue rather than output, $R_j(t) = P(t)\, y_j(t)$ for $\sigma_j(t) \in \{\text{Leasehold}, \text{Freehold}\}$ and $R_j(t) = y_j(t)$ under custom: market-exposed holdings sell their produce at the going price $P(t)$ of §4.14, while customary production is consumed at home and so is not revalued by the urban market. Subsistence is owed *per head*, which is what makes household size a real constraint rather than a free source of labour: output rises with labour as $n_j^{1-\mu-\gamma}$ while this term rises as n_j, so every holding has a size beyond which it cannot feed its own household. (Under the single-body population rules a household is one member and pays the flat c_j of Table 2 instead.) (13) nests both the Leasehold and Freehold cases of earlier drafts as $\rho_j(t) = \hat{r}_i(t)$ and $\rho_j(t) = 0$ respectively. The Leasehold rent is not individually negotiated with the sitting tenant but set once for the whole estate at landlord i's local competitive benchmark – a single going rate a tenant has no part in setting, which is what makes it genuinely competitive rather than a personalised squeeze on whoever happens to be struggling,

$$\hat{y}_i(t) = \frac{1}{|\mathcal{T}_i^L(t)|} \sum_{j \in \mathcal{T}_i^L(t)} y_j(t), \qquad \hat{r}_i(t) = \theta_{\text{rent}}\, P(t)\, \hat{y}_i(t-1), \qquad \theta_{\text{rent}} \in (0,1), \tag{14}$$

(bootstrapped, at a landlord's first-ever conversion, from the estate's own customary rent scale in place of an as-yet-undefined $\hat{y}_i(t-1)$), and struck in money rather than in produce, for the same reason the bid of (11) is: the benchmark is compared against, and paid out of, revenue that has already been priced. It is the going rate against which every Leasehold tenant on estate i is measured, aggregating to the estate's leasehold receipts $R_i^L(t) = \sum_{j \in \mathcal{T}_i^L(t)} \rho_j(t)$ used in $\Delta_i(t)$ (§4.5); no inflation discount is needed here since $\hat{r}_i(t)$ already re-prices with current output, unlike $r_j^C(0)$, which is fixed in nominal terms at grant. This is exactly the calculation Wood attributes to the landlord's own surveyor (§4.7): $\hat{r}_i(t)$ is the "rental value of land...against the actual rents being paid" [Wood, 2002, p.101], made explicit as a number rather than left as a comparison the landlord alone draws.

Improvement as a response to competitive necessity, not a precondition of it. Rather than gating reinvestment on tenure state by fiat, the model makes a tenant's willingness to reinvest an endogenous "improving disposition" $\iota_j^T(t) \in [0,1]$ – the superscript distinguishing it from the landlord's legitimating ideology $\iota_i(t)$ of §4.7, a conceptually distinct variable that happens to answer the same abstract-level question (why does anyone start improving?) for the other class of agent.

What drives it is *competitive selection*, not the level of rent. This distinction is essential and easy to get wrong. The intuitive formalisation – letting the rent-to-output ratio $\rho_j(t)/y_j(t)$ do the work – inverts the theory, because a customary tenant's rent is fixed in nominal terms and so is a *lump sum*: they retain the whole of any extra output they coax from the land, whereas a leaseholder assessed at θ_{rent} of output faces a marginal claim on every additional bushel. On that reading custom would supply the stronger incentive to improve, which is the opposite of Brenner's argument. What actually distinguishes the market-dependent tenant is that they are measured against a rate somebody else sets, and lose the holding for falling short of it. The driver is therefore the estate's competitive benchmark relative to the tenant's own revenue, and it applies only to those tenures whose continued occupation is contingent on meeting it:

$$M_j(t) = \begin{cases} 0 & \sigma_j(t) = \text{Customary}, \\ \dfrac{\hat{r}_i(t)\, |\mathcal{P}_j(t)|}{R_j(t)} & \sigma_j(t) \in \{\text{Leasehold}, \text{Freehold}\}, \end{cases} \qquad \iota_j^T(t{+}1) = (1{-}\varsigma)\, \iota_j^T(t) {+} \beta^T M_j(t) \big(1{-}\iota_j^T(t)\big), \tag{15}$$

both sides of $M_j(t)$ being in money, so that the price carried by $\hat{r}_i(t)$ and by $R_j(t)$ cancels and the pressure is a pure ratio. Here ς is a decay rate: a disposition is sustained by the pressure that produced it, and fades in its absence, so that improvement tracks present circumstances rather than being an acquired trait a dispossessed tenant carries permanently into a tenure that exerts no such pressure. Reinvestment then scales continuously in the disposition, against a depreciating stock,

$$k_j(t+1) = \max\Big(k^{\text{trad}},\ (1 - \varrho_k)\, k_j(t) + \bar{s}\, \iota_j^T(t) \max\big(0,\ w_j(t+1) - w^{\min}\big)\Big), \tag{16}$$

with ϱ_k the depreciation rate. Depreciation matters for more than realism: without it an improvement, once made, is permanent and free to maintain, and the compulsion the theory describes is a one-off rather than a standing condition of continued tenancy.

The other half: why custom does not improve. A zero in (15) states that custom applies no competitive test, but says nothing about whether a customary tenant *could* improve if minded to. Brenner supplies the answer, and the paper has already quoted it (§4.4): seigneurial extraction "tended to confiscate not merely the peasant's income above subsistence...but at the same time threatened the funds necessary to refurbish...productivity" [Brenner, 1976, p.48]. The model implements that directly as an arbitrary fine at rate φ on customary income above subsistence,

$$F_j(t) = \varphi \max\big(0,\ y_j(t) - \rho_j(t) - c_j\big), \qquad \sigma_j(t) = \text{Customary}, \tag{17}$$

deducted from the tenant and credited to the lord's receipts. The two mechanisms are complementary rather than redundant: (15) removes the compulsion to improve under custom, and $F_j(t)$ removes the means. Crediting the fine to the lord also gives the model the choice the theory turns on – a lord under fiscal pressure may squeeze harder instead of converting, so conversion has to out-compete an alternative rather than being the only available response.

Because $\hat{r}_i(t)$ tracks the estate's competitive peers rather than any individual tenant's realised output, the reverse of (15) also holds: a tenant who does not raise $\iota_j^T(t)$ – and hence $k_j(t)$ and $y_j(t)$ – falls behind a rent they had no part in setting, and sustained shortfall triggers the same τ-period eviction rule as rent default generally (below), vacating the parcel for engrossment by a more successful neighbour (§4.10). Reinvestment is therefore not only individually adaptive under pressure but also the criterion by which the tenant population is selected – compete, in the sense of (15), or be outcompeted, in the sense of §4.12. This is the same market compulsion already used to justify Freeholders' continued exposure (§4.6): "even owner-occupiers would be subject to those pressures once the competitive productivity of agrarian capitalism set the terms of economic survival" [Wood, 2002, p.54].

This raises output through a diminishing-returns function of the capital and labour applied across the tenant's whole holding $\mathcal{P}_j(t)$, not a single fixed parcel,

$$y_j(t) = \Phi_j(t)^{\mu} \cdot k_j(t)^{\gamma} \cdot h_j(t)^{1-\mu-\gamma}, \qquad \mu, \gamma \in (0,1),\ \mu + \gamma < 1, \tag{18}$$

where $\Phi_j(t) = \sum_{k \in \mathcal{P}_j(t)} \phi_k(t)$ is the fertility-weighted size of the holding and $h_j(t) = \bar{\ell} + \ell_j^{\text{hired}}(t)$ is total labour applied: the tenant's own household endowment $\bar{\ell}$ plus any hired labour $\ell_j^{\text{hired}}(t)$ drawn from the landless pool (§4.14). How $\mathcal{P}_j(t)$ itself grows is the subject of §4.10; here it is enough that this is the mechanism, not merely an analogy, that Brenner identifies as distinguishing English agrarian development: the displacement of "the traditionally antagonistic relationship in which landlord 'squeezing' undermined tenant initiative, by an emergent landlord-tenant symbiosis which brought mutual co-operation in investment and improvement" [Brenner, 1976, p.65]. A tenant of any tenure state who cannot cover $\rho_j(t) + c_j$ for τ consecutive periods loses the tenancy and enters the Landless state – not to leave the model, but to become the labour supply the surviving tenants in (18) draw on, as §4.14 sets out; this is a crisis-type vacancy in the sense of §4.12, with no heir to fall back on, and is the mechanism by which a tenant who fails to develop $\iota_j^T(t)$ (above) is removed from the model.

4.10 Land concentration: the engrossment of holdings

Neither Brenner nor Wood treats landlord-level estate size as something the sixteenth–seventeenth century dynamics generate: Wood is explicit that big-landlord concentration was already in place before the story starts – "land in England had for a long time been unusually concentrated, with big landlords holding an unusually large proportion, in conditions that enabled them to use their property in new ways" [Wood, 2002, p.99]. Landlord estate size $|\mathcal{T}_i(0)|$ is therefore initialised as a heterogeneous but fixed starting condition (§4.1), not an emergent output. What the theory does explain endogenously is a different margin entirely: the concentration of *farms*, i.e. tenant holdings, as customary smallholdings are merged into fewer, larger units –

> "with the peasants' failure to establish essentially freehold control over the land, the landlords were able to *engross, consolidate and enclose*, to create large farms and lease them to capitalist tenants who could afford to make capitalist investments" [Brenner, 1976, p.63],

a pattern Brenner also documents in the comparative section via Bowden: "under the stimulus of growing population, rising agricultural prices, and mounting land values...large estates were built up at the expense of small holdings" [Brenner, 1976, p.42].

At any vacated parcel k – whether from the ecological-turnover channel of §4.4 or a rent-default eviction (§4.14) – the landlord's decision is therefore not only *whether* to convert (§4.5) but *to whom* to let it: a new, separate tenant, or an existing neighbouring tenant's holding. Writing $\mathcal{N}(k)$ for the Leasehold or Freehold tenants already adjacent to k and $j^* = \arg\max_{j \in \mathcal{N}(k)} k_j(t)$ for the most capital-intensive of them, the probability of engrossment into j^* rather than separate re-letting is

$$q_i(t) = \sigma\big(\psi_0 + \psi_1 \Delta_i(t) + \psi_2\, k_{j^*}(t)\big), \tag{19}$$

reusing the landlord's fiscal pressure $\Delta_i(t)$ (§4.5) and the neighbour's own capital stock $k_j(t)$ (§4.9) rather than introducing new state. If engrossment occurs, $\mathcal{P}_{j^*}(t+1) = \mathcal{P}_{j^*}(t) \cup \{k\}$, directly operationalising Wood's point that "success would breed success, and competitive farmers would have increasing access to even more land, while others lost access altogether" [Wood, 2002, p.100-101]. If not, the parcel enters the ordinary conversion decision of §4.5 as before.

Because the same displaced smallholder who loses the parcel typically becomes the labour the engrossing tenant now needs to work the larger holding (§4.14), the model produces tenant-farm concentration and the wage-labour pool as two faces of a single event, rather than as separately assumed processes.

4.11 Enclosure of the commons

Engrossment operates parcel by parcel, within the individual tenancy relation. Enclosure is a distinct, slower process that operates on shared subsistence resources external to any single tenancy: "there existed common lands, on which members of the community might have grazing rights or the right to collect firewood, and there were various other kinds of use rights on private land, such as the right to collect the leavings of the harvest during specified periods of the year" [Wood, 2002, p.107]. Enclosure is the extinguishing of exactly these rights, not merely the fencing of land: "enclosure meant not simply a physical fencing of land but the extinction of common and customary use rights on which many people depended for their livelihood" [Wood, 2002, p.108], and its early, most disruptive wave targeted pasture for sheep – Thomas More, "though himself an encloser, described the practice as 'sheep devouring men"' [Wood, 2002, p.108-109].

Let $\Xi(t) \in [0,1]$ denote the aggregate share of common/use-right land already enclosed, diffusing alongside improvement ideology – Wood ties the two processes together directly, since it is "reasons of 'improvement"' that were "cited systematically...as grounds for extinguishing traditional rights" [Wood, 2002, p.115] – with a discrete acceleration once state resistance to enclosure disappears:

$$\Xi(t+1) = \Xi(t) + \nu_1\big(1 - \Xi(t)\big) + \nu_2\, \bar{\iota}(t)\big(1 - \Xi(t)\big) + \nu_3 \mathbb{1}[t \geq t^*]\big(1 - \Xi(t)\big), \tag{20}$$

where $\bar{\iota}(t)$ is the population-average of the ideology term $\iota_i(t)$ (§4.7) and t^* is a scenario parameter marking the point at which the state stops contesting enclosure at all: "in its earlier phases, the practice was to some degree resisted by the monarchical state...but once the landed classes had succeeded in shaping the state to their own changing requirements – a success more or less finally consolidated in 1688...there was no further state interference" [Wood, 2002, p.109], the same alignment of state and landlord interest already used to set θ (§4.5) – Brenner independently ties enclosure to the identical alliance, describing the English gentry as the very group "most concerned to undermine peasant property in the interests of enclosure and consolidation" [Brenner, 1976, p.71-72].

Enclosure's effect is external to the tenancy whose rent status changes: it extends the turnover hazard (4) with a third term, raising it for *any* nearby smallholder – Customary tenant or marginal Freeholder alike – who depended on commons access to make ends meet, whether or not their own parcel is the one being converted. A fixed share ρ_{commons} of parcels in each estate is flagged commons-dependent at $t = 0$, chosen at random independently of $\bar{\phi}_k$, and carries $\mathbb{1}[k \text{ commons-dependent}] = 1$ for the life of the run – operationalising Wood's "various other kinds of use rights" (quoted above) as a fixed structural feature of the manor, not one that itself grows or is assigned endogenously,

$$\eta_j^{\text{crisis}}(t) = \eta_1 \max\big(0, \bar{y} - y_j(t)\big) + \eta_2\, \Xi_{i(j)}(t) \cdot \frac{\big|\{k \in \mathcal{P}_j(t) :\ k \text{ commons-dependent}\}\big|}{|\mathcal{P}_j(t)|}, \tag{21}$$

which nests (4) exactly when $\Xi(t) = 0$. A household is exposed to its own lord's enclosure $\Xi_{i(j)}(t)$, and in proportion to the share of its holding that is commons-dependent rather than through a bare indicator; under the national rule of §4.11 every estate carries the same value and this reduces to $\Xi(t)$ exactly, so the distinction bites only under the local rule. This is what lets the model represent enclosure as the specific driver of dispossession *without* an individual tenancy conversion – the historical "growing plague of vagabonds, those dispossessed 'masterless men' who wandered the countryside" [Wood, 2002, p.108-109] – as distinct from, though reinforcing, the rent-conversion and engrossment channels above.

The indicator in (21) is written per parcel, but the hazard is applied per *household*, and once engrossment lets one household hold several parcels the two are no longer the same thing. The implementation therefore uses the *share* of a household's holding that is commons-dependent in place of the indicator, which coincides with (21) for a single-parcel holding and generalises it for a larger one: a household that has engrossed four parcels of which one carried common right is a quarter exposed, not fully exposed. Reported here because it is a departure from the equation as written, and because it makes enclosure's bite fall off as holdings consolidate, which is a substantive consequence rather than a bookkeeping detail.

A national process with a local driver. As specified, $\Xi(t)$ is a single aggregate and $\bar{\iota}(t)$ a population average, so enclosure has a timing but no location. This is faithful to Wood, who describes enclosure as a national movement in its legal and political aspect, but it sits awkwardly with the mechanism it is tied to: the improvement ideology of §4.7 diffuses *locally*, through the awareness neighbourhood, and enclosure is then the only mechanism in the model that is national while its own driver is local. It also confines RQ9 to a claim about timing, when the dispute it addresses – Neeson on the substance and separability of common right [Neeson, 1993] against Shaw-Taylor on its diminished share [Shaw-Taylor, 2012] – is partly a regional one.

The switch `enclosure_rule` therefore admits a variant in which each estate carries its own $\Xi_i(t)$, driven by the mean disposition of that lord and the lords within the awareness radius, with the reported aggregate the parcel-weighted mean. It is not the baseline, and the reason is worth stating rather than burying in a robustness table. England's enclosure geography was largely set by pre-existing field systems – the open-field Midlands were the enclosable land, much of the south-east and west long since enclosed – and the model has no field-system layer. A locally diffusing Ξ_i therefore inherits the geography of conversion, because the two share a driver. **The arm is consequently informative in one direction only.** Two fronts that coincide under it say almost nothing, being close to a construction; two fronts that *diverge* despite the shared driver are a genuine separability result, and one that favours Shaw-Taylor. Reported on those terms in §5, with the national rule as the baseline throughout.

4.12 Vacancy resolution and population continuity

The mechanisms above are all triggered "at a vacancy" without yet specifying, for every vacancy, in what order a landlord's options are exercised, nor where the next occupant of a re-let tenancy comes from. Both questions are answered by distinguishing two kinds of vacancy already implicit in (4): the baseline hazard η_0, representing ordinary generational succession (the death of a tenant with an heir), and the excess hazard $\eta_j^{\text{crisis}}(t)$ – together with rent-default eviction under any tenure state (§4.9), or a tenant's own departure by migration (§4.7) – representing genuine dispossession with no customary claimant left to fall back on. This is exactly the distinction Brenner draws between ordinary conveyance, at which the landlord's only historical lever was the entry fine [Brenner, 1976, p.62], and demographic-collapse vacancies that a landlord "could simply appropriate...and add...to his demesne" [Brenner, 1976, p.61] because no one remained to contest them.

Conveyance vacancies (probability η_0 at any occupied parcel, each period). The tenancy is not structurally vacant: an heir succeeds the outgoing tenant by default, inheriting $\mathcal{P}_j(t)$ and a fraction ξ_{inherit} of their wealth, with capital reset to the traditional baseline k^{trad} (§4.9). This is precisely the moment of "conveyance" at which the landlord may still act (§4.5): the engrossment check and conversion decision below run exactly as at a crisis vacancy, so an entry fine can still be levied on the heir even though the parcel never stood empty.

Since §4.13 gives the household an explicit mortality rate, η_0 must be read strictly as the hazard of *conveyance* – the generational turnover of the head at which the tenancy is regranted – and not as a death rate, or the two would double-count the same deaths and the population would drain at $\eta_0 + m_j(t)$. It is accordingly population-neutral: the heir inherits the household at its current size $n_j(t)$, and the deaths of individual members are handled entirely by (23). The one exception is the option of impartible inheritance (§4.13), under which conveyance passes the holding whole to a single heir and the remaining $n_j(t) - 1$ members leave as a new landless household.

Extinction vacancies. A household that loses its last member (§4.13) leaves its parcels with no claimant of any kind. These are resolved exactly as crisis vacancies below, with the single difference that no dispossessed household joins the landless pool – there is nobody left to dispossess. This is escheat rather than eviction, and it is the mechanism Brenner identifies in the aftermath of demographic collapse, when a landlord "could simply appropriate...and add...to his demesne" [Brenner, 1976, p.61] land that no surviving tenant could contest. Because it takes precedence, an extinct household is never also counted as evicted or as conveyed.

Crisis vacancies. No customary claimant remains, so parcel k genuinely falls to the landlord's disposal. Resolution proceeds in a fixed order:

1. *Engrossment check* (§4.10): draw engrossment with probability $q_i(t)$. If it succeeds, k is absorbed into j^*'s holding and resolution ends here – no new tenant is created.

2. *Conversion decision* (§4.5): if not engrossed, draw conversion with probability $p_i^{\text{convert}}(t)$.

3. *Freehold diversion* (§4.6): if conversion is resisted (the draw in step 2 fails), draw $p_j^{\text{Freehold}}(t)$; on success the new occupant enters directly as Freehold rather than Customary.

4. *New occupant.* If neither engrossed nor converted, the parcel is re-let at its unchanged nominal rent $r_j^C(0)$ – custom attaches to the land, not the person.

   Tenure ratchets. This step applies only where the parcel is *still* customary. Tenure is a property of the parcel and moves in one direction: a leasehold parcel whose tenant fails is re-let as leasehold, never returned to custom. The distinction is not a detail. Read the other way – with every unconverted vacancy reverting to customary tenure regardless of the parcel's history – the customary sector becomes an absorbing state that the model falls back into faster than conversion escapes it, and no transition can accumulate however favourable the conversion decision is. Historically the point is not in doubt: the extinction of customary right is the irreversible process Brenner and Wood are describing, and §4.11's account of use-rights being extinguished rather than suspended says as much. If converted (step 2) or diverted to Freehold (step 3), the new occupant is drawn preferentially from the Landless pool $\mathcal{L}(t)$ (§4.14), subject to an entry cost $\chi \cdot \hat{r}_i(t)$ – the same entry-fine object introduced qualitatively in §4.5, now given a magnitude, where $\hat{r}_i(t)$ is estate i's own competitive rent benchmark (§4.9), the same figure the new tenancy will itself be charged if Leasehold. A Landless agent takes the vacancy if $w_\ell(t) \geq \chi\, \hat{r}_i(t)$, entering with $w_j(0) = w_\ell(t) - \chi\, \hat{r}_i(t)$, $k_j(0) = k^{\text{trad}}$, and $\iota_j^T(0) = 0$.

If step 4 requires a Landless entrant and none clears the entry cost, k joins a vacant-parcel queue, re-attempting steps 1–4 (with $\Delta_i(t)$ and $q_i(t)$ recomputed) every subsequent period until filled; a parcel in this queue produces no output and no rent, itself contributing to $\Delta_i(t)$. Land held by neither a tenant nor the queue does not otherwise exist in the model: every parcel is, at all times, either held, being engrossed, or in the queue.

This closes the model's population accounting without an unmodelled reservoir of prospective tenants: the customary sector reproduces itself through conveyance, matching the historical persistence of the customary peasantry across the period Brenner and Wood describe, while the Leasehold/Freehold sector's growth is bounded by the size of the Landless pool it itself generates through eviction and partition (§4.13, §4.9) – the "reserve army" channel of §4.14 is therefore also, endogenously, the channel that staffs new capitalist tenancies, not merely the wage-labour market. Every occupant of a re-let parcel comes from the Landless pool, so the model never conjures a tenant from nothing, and the cumulative exit share of §4.17(vi) remains a share of a determinate population.

4.13 Household demography: fertility, mortality and partition

Population in this period was not fixed, and an account of agrarian change has to say something about it. The difficulty is that the obvious way to model it is the one Brenner's 1976 intervention was written to reject: an aggregate law in which population grows with the food supply and presses back on the land, so that the outcome of agrarian history is read off a ratio of people to acres. A model in which such a law silently drove the transition would beg the question the paper is asking.

The resolution is to locate the demographic rates at the level at which Brenner locates causation – the individual household's position within a structure of property relations – and let the aggregate rate be whatever the composition of classes makes it. Nothing in what follows references a total population, a food supply, or a carrying capacity. What each household reads is its own per-head surplus relative to its own subsistence requirement,

$$g_j(t) = \begin{cases} \dfrac{y_j(t) - \rho_j(t) - \omega(t)\ell_j^{\text{hired}}(t) - n_j(t)c^{\text{head}}}{n_j(t)\, c^{\text{head}}} & \sigma_j(t) \in \{\text{Customary}, \text{Leasehold}, \text{Freehold}\}, \\[2ex] \dfrac{\omega(t)\bar{\ell} - c_{\ell,j}}{c_{\ell,j}} & \sigma_j(t) = \text{Landless}, \end{cases} \tag{22}$$

a dimensionless quantity, zero when the household exactly meets subsistence. (The arbitrary fine $F_j(t)$ of §4.9 is deliberately excluded, since it is itself levied on customary income above subsistence and would otherwise be charged twice.) The two branches are the substantive point rather than a technicality. A landholding household's demographic prospects are set by its land, its capital and the rent it owes; a landless household's are set by the wage. It follows that the reproduction rate of the population as a whole is determined by the distribution of property – the same land supporting a growing or a shrinking population according to how its surplus is divided – which is Brenner's claim about demography rather than Postan's, and which the model can now express instead of assuming away.

Fertility and mortality. Each member of household j independently gives birth with probability $b_j(t)$ and dies with probability $m_j(t)$,

$$b_j(t) = \min\big(b^{\max},\ m_0 + \beta^b g_j(t)\big), \qquad m_j(t) = m_0 + m_1 \max\big(0,\ -g_j(t)\big), \tag{23}$$

so that fertility is anchored at the baseline mortality m_0 where surplus is zero. A household exactly at subsistence is therefore stationary by construction; prosperity raises fertility to a ceiling $b^{\max}$; and a household in deficit is losing members to *both* sub-replacement fertility and excess mortality, the nutritional channel that m_1 scales. Births and deaths are drawn as Binomial$(n_j(t), b_j(t))$ and Binomial$(n_j(t), m_j(t))$, giving

$$n_j(t+1) = \min\big(n^{\max},\ n_j(t) - \text{deaths} + \text{births}\big), \tag{24}$$

with $n^{\max}$ a numerical guard. A household reaching $n_j = 0$ is extinct: if it holds land, its parcels become the extinction vacancies of §4.12.

Partition: how a household sheds members into the proletariat. A household with more hands than its land can use sends one out to work for wages. The member leaves when they are better placed outside the household than in it, both sides reckoned net of what it costs to keep them: inside, they add the marginal product of labour and eat per-head subsistence; outside, they earn the wage and must meet their own landless requirement. For the production function (18) this gives the condition

$$\underbrace{(1-\mu-\gamma)\frac{y_j(t)}{h_j(t)} - c^{\text{head}}}_{\text{net contribution inside}} < \underbrace{\omega(t)\bar{\ell} - c_{\ell,j}}_{\text{net position outside}}, \tag{25}$$

and a household satisfying it (or in per-head deficit, $g_j(t) < 0$) sheds one member with probability p^{part} per period, the new landless household taking a dowry w^{dowry} from its parent's wealth. Partition applies only to households holding land; a landless household splitting in two would change nothing, since both halves already live by the wage.

Three features of (25) are worth drawing out. It is a comparison the household could actually make from quantities it observes, rather than an accounting threshold imposed on it. It runs the classic peasant response in the right direction: a falling wage makes the right-hand side smaller, so families *retain* members and work the holding harder, while a wage that industry has bid up releases them – the self-exploitation of the family farm and its dissolution being the same mechanism read at two wage levels [Chayanov, 1966]. And because $y_j(t)$ depends on the household's own $\Phi_j(t)$, the productivity of a household's particular land is what sets how many people it can hold: good or consolidated land absorbs population, poor or fragmented land expels it, parcel by parcel and with no aggregate ceiling anywhere in the specification. This is also the channel through which the engrossment of §4.10 feeds back into the supply of wage labour, rather than merely coinciding with it.

Optionally, setting an impartible-inheritance regime makes conveyance itself a partition event: the holding passes whole to one heir and the remaining $n_j(t) - 1$ members leave together as a new landless household. This is the strong form of the same pump, and it is off by default because (25) already generates the flow without having to assume a uniform inheritance custom; turning it on asks how much of the transition the inheritance regime alone can drive.

Why this replaced a closed population. Earlier drafts held the population fixed, creating nobody after $t = 0$, on the reasoning above that a model should not be allowed to explain the transition demographically. The specification was defensible but the implementation was not, and the reason is instructive enough to record. With no births and exit to industry absorbing, the headcount can only fall: $N(t) = N(0) - \text{exits}(t)$ identically. Over a 200-period run this cost 73% of the population, and by $t \approx 120$ an average of 1,347 of 7,400 parcels stood in the vacancy queue of §4.12 because no living agent remained to take them up. Departures also arrived in cohorts rather than as a flow: because the wage is global and every landless agent faced the same subsistence requirement and the same integer counter, 22% of all exits occurred within five periods, after which the drained pool drove the wage to two and a half times subsistence and left it there. The second half of every run therefore conducted its consolidation under labour scarcity and high wages – the inverse of the conditions the theory describes, in which enclosure coincides with a growing propertyless mass and cheap labour [Wood, 2002, p.103]. Across seeds the final Leasehold share correlated at $+0.97$ with how far the population had fallen, which makes the headline result a statement about the population artifact rather than about property relations.

The population is accordingly conserved in the only sense that a demographic model should conserve it: the *total* head count, rural plus urban, changes through births and deaths alone, while exit to industry ((30)), return from it ((31)) and partition ((25)) are transfers between positions and alter no total. Both identities are asserted every period as a runtime invariant. This is what makes the reported series interpretable: a falling rural population can now be attributed to migration or to natural decrease, which are different claims, where previously the two were indistinguishable because only one of them was possible.

A closed population is retained as a distinct arm, because "does the transition still occur with population held fixed?" remains a real test of Brenner's claim – RQ7, and a claim about the theory rather than a robustness check on the model (§3.1) – and it should be reported as such. It cannot be the default. That English population in fact grew substantially across this period is not in dispute [Wrigley and Schofield, 1981], and Brenner himself invokes population growth as a condition under which "large estates were built up at the expense of small holdings" [Brenner, 1976, p.42]; what he denies is that it *explains* the divergence between England and France, both of which experienced it. Equations (22)–(25) are constructed to respect exactly that distinction: population responds to the class structure, and never appears as a force acting on it.

4.14 The wage-labour pool: hiring, wages, and exit

A landless agent is not a sink. Brenner's consolidation mechanism only works because the larger holdings it produces need more hands to farm than a single household supplies: demesne land was "leased out to larger tenants...[who] in turn sub-let to smaller farmers or employed wage labourers" [Brenner, 1976, p.61]. The model must therefore specify what the landless do next, not merely how tenants come to lose their land.

Labour demand. A Leasehold or Freehold household hires up to the point at which the marginal product of labour equals the going wage, net of the labour its own members already supply, which for the production function (18) gives

$$\ell_j^{\text{demand}}(t) = \max\Big(0,\ \Big[\tfrac{(1-\mu-\gamma)\,\Phi_j(t)^{\mu} k_j(t)^{\gamma}}{\omega(t)}\Big]^{\frac{1}{\mu+\gamma}} - n_j(t)\,\bar{\ell}\Big), \tag{26}$$

so that it is precisely the successful, improving, consolidating tenants – Wood's "success would breed success" [Wood, 2002, p.100-101] – who hire, and unsuccessful ones who are hired. Family and hired labour are thus substitutes at the margin, and because only the hired part is charged at $\omega(t)$ in (13), they are not substitutes at equal cost: a large household working its own holding undercuts an equally large holding staffed by wage labour, which is the family farm's advantage and the reason (25) rather than a wealth threshold governs when it gives a member up. Demand has to reference $\omega(t)$: a rule in capital alone leaves nothing in the model responding to the price of labour, so the tâtonnement below has no equilibrium to find and the wage simply diverges. Correspondingly, the hiring tenant pays the wage bill $\omega(t)\ell_j^{\text{hired}}(t)$, which enters (13) alongside rent and subsistence; without it hired labour is free, and capital, output and wealth compound on one another without limit. Demand is a claim on a finite pool, not a guarantee: if aggregate demand $D(t) = \sum_j \ell_j^{\text{demand}}(t)$ exceeds the pool, hiring is rationed pro rata,

$$\ell_j^{\text{hired}}(t) = \ell_j^{\text{demand}}(t) \cdot \min\Big(1,\ \frac{S(t)}{D(t)}\Big), \qquad S(t) = \sum_{\ell \in \mathcal{L}(t)} n_\ell(t)\,\bar{\ell}, \tag{27}$$

so that $h_j(t) = n_j(t)\bar{\ell} + \ell_j^{\text{hired}}(t)$ in (18) never draws on more bodies than actually exist. Note that the supply $S(t)$ counts *bodies*, not households: a landless household of four supplies four labour endowments even though it bids once at an auction (§4.8). The unmet portion of demand – rather than a phantom labour supply – is exactly what the wage adjustment below prices.

Wage determination. The same excess demand $D(t) - S(t)$ that rationing absorbs in quantity is what moves price, through a simple tâtonnement,

$$\omega(t+1) = \omega(t)\Big(1 + \kappa\,\frac{D(t) - S(t)}{S(t)}\Big), \tag{28}$$

so that a labour supply growing faster than tenant demand for hired labour depresses the wage – the model's "reserve army" channel, consistent with the historical pattern of rents rising while wages stayed comparatively flat over the same period [Kerridge, 1953, Clark, 2002]. Because $S(t)$ is fed by the partition rule of §4.13 as well as by eviction, the wage is where the two halves of the theory meet: dispossession and household formation both price labour, and the wage in turn feeds back into (25) to govern how readily the next household gives a member up.

Landless income and exit. A landless household sells the labour endowment of every member at the going wage and feeds every member out of the proceeds,

$$w_j(t+1) = w_j(t) + n_j(t)\big(\omega(t)\,\bar{\ell} - c_{\ell,j}\big), \tag{29}$$

so that its per-head position, and hence its fate, is independent of its size. The requirement $c_{\ell,j}$ is drawn per household from $\mathcal{U}\big(c_\ell(1-\varepsilon_c),\, c_\ell(1+\varepsilon_c)\big)$ and fixed for its lifetime. This matters more than it appears to. With a single shared requirement, and a wage that is by construction global, every landless household in the model faces an identical net income, crosses zero on the same tick, and – under a deterministic counter – leaves on the same tick as well; the cohort behaviour described in §4.13 is the consequence. Departure is correspondingly a hazard in the *depth* of a household's deficit rather than a threshold in its duration,

$$p_j^{\text{exit}}(t) = 1 - \exp\Big(-\lambda^{\text{exit}} \frac{\max(0, -w_j(t))}{n_j(t)\, c_{\ell,j}}\Big), \tag{30}$$

measured in periods of the household's own consumption, so that a household barely under water rarely leaves and one deeply in debt almost surely does, and two otherwise identical households leave at different times. Exit is not a modelling convenience but the mechanism Brenner names directly: "agricultural improvement not only made it possible for an ever greater proportion of the population to leave the land to enter industry" [Brenner, 1976, p.67]. In the reverse direction, a landless household whose wealth clears the entry cost $\chi \cdot \hat{r}_i(t)$ of an open vacancy (§4.12) may re-enter directly as a Leasehold tenant – or, at a conveyance vacancy left without an heir, as a Customary one – giving the model a (theoretically vanishing, as land concentrates) channel of upward mobility rather than assuming the triad, once formed, is a closed caste system. The older deterministic rule, exit once $w_j(t) < 0$ for τ' consecutive periods with $\varepsilon_c = 0$, is retained as a switch for comparison.

The urban sector is a place, not a sink. Leaving for industry is a move, and treating it as terminal is not the neutral choice it appears to be. An absorbing urban state builds a one-way ratchet into the population accounting: the countryside can then only lose people to it, whatever its own vital rates, and the resulting decline is reported as though it were a finding. In the runs behind §4.13 this was quantitatively decisive rather than a nicety – rural births exceeded rural deaths by 11,256 over 200 periods while 20,369 persons left for industry and none came back, so a countryside that was reproducing perfectly well still ended at 58% of its initial population, for no reason other than the valve.

An urban household therefore persists with its size and wealth, earns ω^U per member, pays c^{urban} per member, and is subject to the same vital rates (23) as everyone else on the per-head surplus those two imply. It returns to the landless pool with probability

$$p_j^{\text{return}}(t) = p^{\text{ret}} \cdot \min\Big(1,\ \max\Big(0,\ \frac{\big(\omega(t)\bar{\ell} - c_{\ell,j}\big) - \big(\omega^U - c^{\text{urban}}\big)}{c_{\ell,j}}\Big)\Big), \tag{31}$$

the same net-position comparison that governs partition (25), read in the opposite direction. It is scaled by the *size* of the advantage rather than triggered by its sign, because returning households enlarge the labour pool and so depress the very wage that drew them: on a bare sign test that feedback produces a churn loop, cycling people between town and country on a vanishing differential, rather than a response to conditions.

Two calibration points follow, and both are consequential enough to state rather than bury. ω^U is exogenous – the model has no industrial sector to derive it from, and inventing one would extend the paper well past the agrarian transition it is testing – so holding it fixed while $\omega(t)$ moves endogenously is what makes the direction of migration respond to conditions in the *countryside*, which is the causal direction the theory is about. And it is set equal to c^{urban}, making the urban per-head surplus exactly zero and, by the anchoring of (23) at m_0, the urban sector demographically stationary. That is the neutrality the channel is for: the towns neither breed nor consume a population of their own, so every movement in the rural/urban split is migration answering to rural conditions. Departing from equality is a substantive claim about urban demography that this model has no evidence for, and it is not a small one – at $\omega^U = 0.9$ against $c^{\text{urban}} = 1$ the towns lose about 1.9% of their people per period, which drained 74% of the *total* population over 200 periods and put a third of all parcels back into the vacancy queue.

The domestic market. Nor do urban households only supply labour: they are the market the remaining farms sell into. Wood's claim is that agrarian productivity produced "an increasing propertyless mass that would constitute both a large wage-labour force *and* a domestic market for cheap consumer goods" [Wood, 2002, p.103], and without the second half the exit channel is a pure drain in which the countryside never feels the consequences of its own dispossession. Writing $U(t)$ for the number of persons currently in the urban sector – a stock, since (31) lets them leave it – the produce price adjusts to the excess of their demand over what the market-exposed sector actually markets,

$$P(t+1) = P(t)\Big(1 + \kappa_P\,\frac{U(t)\,c^{\text{urban}} - Y^M(t)}{Y^M(t)}\Big), \qquad Y^M(t) = \sum_{j:\,\sigma_j\in\{\text{L,F}\}} y_j(t), \tag{32}$$

and market-exposed output is valued at $P(t)$ in (13) and (22). Customary production is not: it is largely eaten at home, so it is not revalued by a market the household does not sell into – which is itself a difference in kind between the tenures, and one that widens as $U(t)$ grows.

The point of specifying this rather than leaving "wage labourer" as a terminal label is that the theory's own account of *why* the transition matters beyond agriculture runs through exactly this population: rising agricultural labour-productivity is what "created a highly productive agriculture capable of sustaining a large population not engaged in agricultural production, but also an increasing propertyless mass that would constitute both a large wage-labour force and a domestic market for cheap consumer goods – a type of market with no historical precedent. This is the background to the formation of English industrial capitalism" [Wood, 2002, p.103]. The exit channel above is what lets the model, in principle, speak to that claim rather than only to the agrarian triad in isolation.

4.15 Model schedule

The mechanisms above are specified as update rules for individual state variables; what has not yet been fixed is the order in which they run within a single period $t \to t+1$, nor the bookkeeping state – beyond tenure, holdings, wealth and capital – needed to evaluate them. Table 1 lists every piece of state actually required, including the consecutive-shortfall counter $d_j(t)$ that the τ-period eviction rule (§4.9) presupposes but does not itself name. The list below then fixes the order. All quantities dated t (rents, fiscal pressure, the observed-outcomes and ideology terms, wage) are computed once at the start of the period from state as of t and held fixed while every within-period event is resolved – a standard synchronous-update convention, adopted so that two vacancies arising on the same estate in the same period are resolved against the same $\Delta_i(t)$ rather than one seeing a value the other has already changed.

Agent	State	Meaning
Landlord i	$W_i(t)$, $\iota_i(t)$	wealth; legitimating ideology exposure (§4.7)
	$\mathcal{T}_i(t)$	set of currently active, separately-let tenancy slots on the estate
Tenant j	$\sigma_j(t)$, $\mathcal{P}_j(t)$	tenure state; set of held parcels
	$w_j(t)$, $k_j(t)$, $\iota_j^T(t)$	wealth; capital; improving disposition (§4.9)
	$n_j(t)$	household size in adult labour equivalents (§4.13)
	$c_{\ell,j}$	own landless subsistence requirement, drawn once (§4.14)
	$d_j(t)$	consecutive periods $w_j(t)$ has been short of $\rho_j(t)+c_j$; eviction at $d_j(t)=\tau$
Landless ℓ	$w_\ell(t)$, $n_\ell(t)$	wealth; size, supplying $n_\ell(t)\bar{\ell}$ to the labour pool
Urban u	$w_u(t)$, $n_u(t)$	wealth; size. Still a household: it earns, eats, reproduces and may return (§4.14)
Parcel k	$\phi_k(t)$, $\bar{\phi}_k$	current fertility stock; fixed carrying capacity (§4.4)
	commons-flag	fixed at $t=0$ (§4.11)
Global	$\omega(t)$, $P(t)$, $\Pi(t)$, $\Xi(t)$	wage; produce price; price index; enclosure share

Table 1: State carried by each agent type, in addition to the fixed per-agent parameters of Table 2.

1. **Exogenous update.** Draw $\pi(t)$ and update $\Pi(t)$; draw $\epsilon_k(t)$ for every parcel. Since $\phi_k(t+1)$ ((3)) depends only on $\phi_k(t)$ and $\epsilon_k(t)$, it may be computed at any point in this list. Update the produce price $P(t+1)$ against urban demand (§4.14).

2. **Labour demand and clearing.** For every Leasehold or Freehold tenant, compute $\ell_j^{\text{demand}}(t)$ from $k_j(t)$ and $n_j(t)$; ration to $\ell_j^{\text{hired}}(t)$ against the labour supply $S(t)$; update $\omega(t+1)$ (§4.14).

3. **Production.** Compute $y_j(t)$ ((18)) for every occupied tenancy from $\Phi_j(t)$, $k_j(t)$ and $h_j(t) = n_j(t)\bar{\ell} + \ell_j^{\text{hired}}(t)$.

4. **Wealth, capital and disposition.** With $\rho_j(t)$ already fixed at the start of t (Customary and Leasehold rents were set at $t-1$; Freehold is always 0), update $w_j(t+1)$ ((13)), $M_j(t)$ and $\iota_j^T(t+1)$ ((15)), and $k_j(t+1)$; update $d_j(t+1)$ (reset to 0 if solvent, incremented otherwise). Landless households update $w_j(t+1)$ symmetrically and draw $p_j^{\text{exit}}(t)$ ((30)); urban households update theirs at $\omega^U - c^{\text{urban}}$ per member and draw $p_j^{\text{return}}(t)$ ((31)). Record every household's per-head surplus $g_j(t)$ ((22)) for the next step.

5. **Demography.** For every surviving household, urban ones included, draw deaths, then births, from (23) and update $n_j(t+1)$; mark households reaching $n_j = 0$ extinct; then draw partition ((25)) and create the resulting landless households (§4.13). Households created in this step take no part in it, so a household cannot be founded and reproduce in the same period.

6. **Landlord fiscal pressure.** Compute $R_i^C(t)$, $R_i^L(t) = |\mathcal{T}_i^L(t)|\,\hat{r}_i(t)$, and $\Delta_i(t)$ (§4.5).

7. **Spread terms.** Compute $\bar{I}_i(t)$ from $s_{i'}^L(t)$ and $W_{i'}(t)$ as of the start of t (§4.7); update landlord ideology $\iota_i(t+1)$ and effective resistance $\theta_i^{\text{eff}}(t)$; update enclosure share $\Xi(t+1)$ (§4.11).

8. **Vacancy determination.** For every occupied parcel, draw a succession event (probability η_0) or a crisis event (the excess-hazard terms of (21), or $d_j(t+1) = \tau$, or a successful migration draw $p_j^{\text{migrate}}(t)$, §4.7); collect the resulting list of vacancies (§4.12).

9. **Vacancy resolution.** Resolve every vacancy collected in step 7, in random order, by the fixed procedure of §4.12, using the period-t values of $\Delta_i(t)$, $q_i(t)$, $p_i^{\text{convert}}(t)$ and $p_j^{\text{Freehold}}(t)$ computed above – *not* recomputed between vacancies within the same period.

10. **Rent reset.** Recompute $\hat{y}_i(t)$ (§4.9) from this period's realised Leasehold output (step 3) over $\mathcal{T}_i^L(t)$ as it stood during production, giving $\hat{r}_i(t+1) = \theta_{\text{rent}}\hat{y}_i(t)$ for the next period.

11. Advance to $t+1$ with all updated state now current.

4.16 Parameters and initial conditions

Table 2 collects every free quantity introduced above with a placeholder starting value. Because §4.17 treats the model's outputs as a qualitative plausibility check rather than a fitted calibration target, these are deliberately round starting points for the first runs and the sensitivity sweeps that follow, not estimates drawn from data.

Symbol	Value	Meaning
Space and population (§4.2)		
L	155	lattice resolution: L cells span the longer axis of England's bounding box and the shorter axis takes as many square cells of the same size as it holds, giving a 155×136 grid of which 7,417 cells survive the boundary mask
N_L	10 per county	landlords/estates, seeded within each historic county so that no estate straddles a county boundary; 39 counties give 390 estates
r	5	landlord awareness radius, in lattice cells
ζ	2	within-county fertility jitter (§4.2)
Initial agent state (§4.1)		
$W_i(0)$	100	landlord initial wealth
C_i	$\mathcal{U}(0.85, 1.05)\times$ rent roll	consumption requirement, scaled to the estate (§4.5)
initial σ_j shares	0.95 / 0.05 / 0 / 0	Customary / Freehold / Leasehold / Landless
$w_j(0)$	10	tenant initial wealth
k^{trad}	1	traditional (non-market) capital baseline
$r^C_k(0)$	$\mathcal{U}(0.5, 1.5)$	customary rent at grant, drawn per *parcel*
$\bar{\ell}$	1	labour endowment *per household member*
$n_j(0)$	3	initial household size (§4.13)
μ_j	$\mathcal{U}(0, 0.3)$	tenant mobility propensity
Exogenous processes (§4.4)		
$\pi(t)$	0.02	inflation rate per period
ρ	0.1	fertility regrowth rate
$\bar{\phi}_k$	county baseline $\pm\zeta$ (§4.2)	carrying capacity, from ALC grades [Natural England, 2021]
$\epsilon_k(t)$	Bernoulli(0.05) $\times\, \mathcal{U}(0, 2)$	ecological shock
$\bar{y}$	2	subsistence output threshold
η_0, η_1, η_2	0.02, 0.05, 0.03	turnover hazard coefficients
Conversion, resistance and Freehold (§4.5, §4.6)		
$\alpha_0, \alpha_1, \alpha_2, \alpha_3$	-4, 2.5, 3, 1	conversion logit coefficients; pressure necessary, not sufficient
ϖ	0.05	hazard of reopening terms on a sitting tenant (§4.5)
θ	2 (England) / 6 (France)	state–landlord alliance regime
ξ_0, ξ_1	0.3, 0.5	ideology discount on resistance
$\lambda_0, \lambda_1, \lambda_2$	-5, 0.3, 1	Freehold-diversion logit coefficients
t^*	100	period at which state resistance ends
χ	2	entry cost, in periods of rent
ξ_{inherit}	0.5	wealth fraction retained at succession
Spread (§4.7)		
β_1, β_2	0.002, 0.15	ideology diffusion: autonomous, and from neighbour conversions
$\iota_i(0)$	0	initial ideology exposure
Tenant economy (§4.9)		
θ_{rent}	0.3	competitive-benchmark extraction share, $\hat{r}_i(t) = \theta_{\text{rent}}\hat{y}_i(t-1)$
β^T	0.1	necessity-driven growth rate of tenant improving disposition $\iota^T_j(t)$
ς	0.02	decay of the improving disposition absent competitive pressure
φ	0.5	arbitrary-fine rate on customary income above subsistence
ϱ_k	0.05	capital depreciation
$\iota^T_j(0)$	0	initial tenant improving disposition (all tenure states)
$\bar{s}$	0.3	reinvestment-rate ceiling, scaled by $\iota^T_j(t)$
$w^{\min}$	0	wealth floor below which no reinvestment occurs
c_j	1	tenant subsistence requirement, per household (single-body population rules only; the household rule charges $n_j(t)\,c^{\text{head}}$ instead)
μ, γ	0.3, 0.3	Cobb–Douglas fertility/capital exponents
τ	3	periods of default tolerated before eviction
Engrossment and enclosure (§4.10, §4.11)		
ψ_0, ψ_1, ψ_2	-2, 3, 0.1	engrossment logit coefficients
ρ_{commons}	0.2	share of parcels flagged commons-dependent
ν_1, ν_2, ν_3	0.01, 0.05, 0.1	enclosure diffusion coefficients
$\Xi(0)$	0	initial enclosure share
Labour market and the domestic market (§4.14)		
κ	0.1	wage adjustment speed
$\omega(0)$	1	initial wage
c_ℓ	0.8	mean landless subsistence requirement, per head
ε_c	0.25	half-width of the per-household draw on $c_{\ell,j}$
λ^{exit}	0.35	exit-hazard scale in (30)
τ'	3	periods of negative wealth before exit (deterministic rule only)
c^{urban}	1	food demanded, and subsistence owed, per urban head
ω^U	1	urban wage per head; set to c^{urban} so the towns are

Two of the demographic values are exceptions to the "round starting point" rule above, in that the model is qualitatively wrong at the obvious choice, and it is worth saying why rather than leaving them to look arbitrary.

$n_j(0) = 3$ rather than 1. Once mortality applies per member, a household of one is extinguished by a single death: at $m_0 = 0.02$ that is a 91% chance of the line failing within 120 periods. Runs started from single bodies lose roughly half their households to extinction before the transition is under way, and the parcels those households held cannot be re-let because the labour pool is not large enough to supply takers – reproducing, by a different route, the vacancy pathology that §4.13 records for the closed population. A household line is not as fragile as a person, and starting from an established family is what makes that distinction operative.

$c^{\text{head}} = 0.5$ rather than the single-body subsistence $c_j = 1$. This constant fixes where partition bites, and therefore how much of the model's proletarianisation is demographic at all. Output rises with labour as $n_j^{1-\mu-\gamma}$ while subsistence rises as n_j, so a household on one parcel becomes unviable at roughly $n_j = (\bar{y}^{\text{parcel}}/c^{\text{head}})^{1/(\mu+\gamma)}$; against a mean output near 1.8 per parcel, 0.5 places that threshold at four to five members, giving mean household sizes of two to three, an extinction rate of a fraction of a percent per period, and mild aggregate population growth. Raising it toward 0.7 drives households back toward single bodies and multiplies extinctions – the failure mode the household unit exists to avoid; lowering it toward 0.45 produces markedly faster growth. It is reported here as a calibrated quantity, and belongs in the sensitivity analysis accordingly.

4.17 Emergent outputs and validation targets

Because the model is a validity test rather than an illustration, its outputs are defined as things that could fail to match the theory's claims. At minimum: (i) the share of tenancies in each tenure state over time, now including Freehold alongside Customary, Leasehold and Landless; (ii) *farm*-size concentration – the Gini coefficient of tenant holding size $A_j(t) = |\mathcal{P}_j(t)|$, which the engrossment mechanism of §4.10 predicts should rise over the run – kept *explicitly distinct* from landlord-estate-size concentration, which §4.10 treats as a fixed initial condition rather than an emergent target, so the model is not credited with reproducing a claim the theory itself does not make; (iii) whether conversion spreads with a distance-dependent lag or occurs simultaneously – the direct test of the "localised seed" claim, measured as the semivariogram of first-conversion time over the lattice, which asks whether parcels further apart convert further apart in time without requiring any point to be nominated as the origin of the transition; (iv) aggregate output $Y(t) = \sum_j y_j(t)$, compared *qualitatively* against the historical rent, wage and grain-output series assembled from secondary sources [Clark, 1991, Clark, 2002, Kerridge, 1953], used as a plausibility check on the shape of the transition rather than as a calibration target; (v) a direct comparative test of Brenner's own argument, by re-running the model with θ set to a "France" regime (institutional resistance systematically prevails), checking not only whether the Leasehold triad fails to emerge but whether the same displaced pressure instead resolves into entrenched Freehold smallholding (§4.6), as the theory predicts: "in England, it was precisely the absence of such rights that facilitated the onset of real economic development" [Brenner, 1976, p.74-75]; (vi) the composition of the landless pool over time – the share still seeking agricultural hire, the wage series $\omega(t)$ against the historical rent/wage ratio, and the cumulative share that has exited the rural model – as the model's operationalisation of the claim that agrarian improvement is what freed population "to leave the land to enter industry" [Brenner, 1976, p.67]; (vii) the enclosure share $\Xi(t)$ against the timing of dispossession implied by tenure-state shares, testing whether the model reproduces enclosure and rent-conversion as related but empirically separable drivers of landlessness, rather than a single conflated process; and (viii) the population as a *composition* – the share of all living persons who are Customary, Leasehold, Freehold, Landless or urban, against a total that is itself an outcome (§4.13). The composition is the target rather than the rural head count, for two reasons. Read alone, a falling rural population cannot distinguish a countryside losing people to towns from one failing to reproduce, and the theory makes a claim about the first and no claim about the second. And the tenure shares of (i) are shares of *tenancies*, so they are silent on how many people stand behind each one; a Leasehold sector that has consolidated onto larger farms worked by larger households is a different outcome from one that has not, and only a per-person denominator shows it. The qualitative pattern the theory implies is a rising total, a rising urban share, and a rural share whose *internal* composition shifts from Customary toward Leasehold and Landless – and it is a real possibility that the model produces the tenure shift without the population shift, which would be a negative result about the sufficiency of the mechanisms rather than a technical failure.

4.18 The historical series used as plausibility checks

Targets (iv) and (vi) above are read against four secondary series, listed in Table 3 so that what is being compared against what is visible rather than buried in a citation. Kerridge covers the model's own window; the Clark series are longer and are used for the shape of the movement rather than for its level.

Source	Series	Period	What it is read against
Kerridge [Kerridge, 1953]	Rent	1540–1640	$\hat{r}(t)$ and the mean real customary rent; the only series covering the model's own window
Clark [Clark, 2002]	Land rental values	1500–1914	The same, over a longer run, for the shape of the movement
Clark [Clark, 1991]	Yields per acre	1250–1860	Aggregate output $Y(t)$ and output per parcel
Clark [Clark, 2004]	Prices and wages	1209–1914	The wage series $\omega(t)$, and the rent/wage ratio of target (vi)

Table 3: The secondary series behind validation targets (iv) and (vi). Read for direction and relative timing, not for level: §4.16 disclaims calibration, so a mismatch in level is not a failure and a mismatch in direction is.

Why these three quantities together rather than any one of them. Brenner's landlord–tenant symbiosis predicts that rents and output rise *jointly* – both classes accumulating – while wages lag both; continental squeezing is the pattern in which rent rises and output does not [Brenner, 1976, p.48, p.65]. The triple is therefore the empirical shape that separates the two cases, and no single series does. This is also the comparison RQ8 makes internal to the model by sweeping θ_{rent}, so the historical panel and that sweep are two views of one claim.

5 Results

Results are organised by research question rather than by figure, and every figure below is produced by one named scenario or comparison of scenarios from the ladder of §5.1. Two conventions hold throughout. First, every series is a mean across seeds with a 95% confidence interval on that mean, because every mechanism in the model is stochastic and a single run cannot distinguish a mechanism from a lucky draw; where the interval on the mean is narrow while individual runs diverge, the per-seed panel is reported alongside it, since a tight interval on a bimodal distribution is a statement about arithmetic rather than about the model. Second, an ablation is always reported against the same seeds as its baseline, so that a difference between arms is not a difference between draws.

5.1 The complexity ladder

The mechanisms of §4 interact, and a run with all of them switched on cannot say which is responsible for what. The scenarios of `model_scenarios/scenarios.yaml` therefore build the model up in tiers, each adding one mechanism to the one before, so that a dynamic can be attributed to the tier at which it first appears. Tier 0 is the bare exogenous skeleton – inflation eroding fixed customary rents against no conversion at all – and each subsequent tier restores one element: the conversion decision, then the spread channels, then engrossment and competitive allocation, then enclosure, then the labour market and the urban sector, then endogenous demography. Every tier runs the full multi-seed replication. The ladder is reported once, as Figure 2, and thereafter treated as the reference against which the question-specific arms are read.

Figure 2: The complexity ladder. Each panel adds one mechanism to the tier before it; the series shown are the Leasehold share, the farm-size Gini, the landless share and aggregate output. A dynamic first appearing at tier n is attributable to the mechanism tier n introduces, which is what a single all-mechanisms run cannot establish.

5.2 RQ1: competitive tenancy or secure property

Figure 3: Improving disposition $\iota_j^T(t)$ and capital $k_j(t)$ by tenure state, Leasehold against Freehold on the same land in the same runs. Brenner's landlord–tenant symbiosis predicts Leasehold leads; Allen's yeoman predicts Freehold does. The right-hand panel repeats the comparison with `freehold_shadow_rent` disabled, which removes the competitive yardstick from Freehold entirely: the two panels bracket the question rather than answering it, and the interval between them is the reported result.

Figure 4: The within-tenure half of RQ1: every converted parcel aligned on its own conversion date, against parcels that never converted. Reported with the occupant-continuity diagnostic, since a conversion that replaces the sitting tenant records a change of occupant rather than a change of disposition (§4.12).

5.3 RQ2: spreading or simultaneous

Figure 5: The distance-lag slope under each combination of the two transmission channels. A positive slope is contagion: conversion reaches distant parcels later. A slope near zero under the "neither" arm, with conversion still completing, is the finding that Brenner's stated mechanisms produce simultaneity and that the account requires an inter-estate channel it does not supply.

Figure 6: The semivariogram of first-conversion time for the full-channel model: half the mean squared difference in conversion time between pairs of parcels, against the lattice distance separating them, normalised by the total variance. No origin is assumed. A curve rising from a low value at short distance is a spreading front – neighbours convert together, distant parcels do not – and the distance at which it flattens is the spatial scale of the process. A flat line at unity throughout is simultaneity: parcels one cell apart convert as differently as parcels a country apart. The map panel shows conversion time in space.

5.4 RQ3: necessary, sufficient or permissive

Figure 7: The 2×2 of RQ3: England and France values of θ crossed with the class-conflict mechanism enabled and disabled ($\alpha_1 = 0$, $\varphi = 0$). The England outcome surviving the disabled column would make the political condition sufficient on its own, which is Bois's charge; θ merely rescaling a transition that occurs in both columns would make it permissive.

5.5 RQ4: how secure custom could have been

Figure 8: The frontier of RQ4. Final Leasehold share over the sweep of the terms-reopening hazard ϖ, the arbitrary-fine rate φ and institutional resistance θ, with the contour at which the transition fails to complete within the run. The reported result is a bound on how secure customary tenure could have been while England still arrived, which is the question Whittle's and Bailey's evidence raises and cannot settle.

5.6 RQ5: necessity of dispossession

Figure 9: Conversion routed through succession vacancies and fine escalation alone, with the crisis channel damped ($\eta_1 = \eta_2 = 0$, eviction suspended), against the full baseline. Leasehold dominance, concentration and a wage-labour pool surviving the damped arm would make dispossession sufficient but not necessary, reconciling Brenner's mechanism with Whittle's Norfolk evidence.

5.7 RQ6: ecology or class structure

Figure 10: The ALC-derived regional fertility structure flattened to its national mean (`uniform_fertility`), retaining parcel-level jitter, against the baseline. Because the distance regressor still measures from the same county, a distance lag surviving this ablation is not an ecological one; a lag that disappears with it is Moore's point established using Brenner's own mechanisms.

5.8 RQ7: does demographic pressure suffice

Figure 11: The transition under a closed population, under vital rates applied to single-body households, and under the full household demography of §4.13; and, in the right-hand panel, under endogenous births with the class mechanism damped as in RQ3. A transition proceeding with population fixed is Brenner's anti-demographic claim; one requiring endogenous births is Postan's.

5.9 RQ8: the edges of symbiosis

Figure 12: Landlord and tenant accumulation across the sweep of the extraction share θ_{rent}. A band in which both rise is Brenner's English symbiosis; its upper edge, beyond which extraction destroys the tenant's capacity to improve and with it the lord's rent roll, is the boundary Brenner asserts without locating.

5.10 RQ9: enclosure as driver or consequence

Figure 13: The landless share and its composition with enclosure suppressed ($\nu_1 = \nu_2 = \nu_3 = 0$) against the baseline, and the timing of $\Xi(t)$ against the timing of rent-conversion-driven dispossession. A landless pool of comparable size without enclosure supports Shaw-Taylor against Wood on the separability of the two channels.

Figure 14: The same question asked of geography rather than of timing, under the local enclosure variant of §4.11: the front of rent conversion beside the front of enclosure, and the two against each other per parcel. The asymmetry of §4.11 governs how this is read – the two processes share a driver and enclosure has no independent spatial input, so fronts that travel together are close to a construction, while fronts that separate are a result. Under the baseline national rule the right-hand panels are empty by construction, and the figure says so.

5.11 Claims about the model

Figure 15: The diagnostics of §3.2: aggregate tenure composition, the attribution of the farm-size Gini to engrossment rather than turnover, the wage and rent series against each other, and the population accounting – the share of parcels no household can take up, and the cross-seed correlation between tenure outcomes and population. A failure in any panel is a defect to repair before the results above can be read.

6 Discussion

7 Conclusions

8 Acknowledgements

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